Research Report

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The Effectiveness of U.S. Economic Policies Regarding China Pursued from 2017 to 2024

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About This Report

In this report, we examine the effectiveness of U.S. economic policies regarding China, focusing on those pursued from 2017 to 2024. We begin with a summary of two of the United States' primary economic policy goals regarding China: promoting fair trade and defending U.S. interests. Using this framework, we analyze the effectiveness of policies regarding tariffs, restrictions on investment and exports, supply chain diversification, efforts to boost U.S. production and manufacturing of key products and materials, and other measures. We also examine China's response to these policies and assessments by the Chinese government and commentators regarding their effects. We use the research conducted for this report to make assessments regarding the effectiveness of U.S. policies toward China related to trade, controls on technology, economic diplomacy, foreign investment, industry, and diversification of supply chains away from China. Finally, we offer recommendations for revising and implementing new economic policies toward China to better achieve U.S. policy goals.

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Summary

The Biden administration and previous U.S. presidential administrations have adopted much more restrictive policies concerning trade, investment, and flows of technology between China and the United States than in the past. The United States has also passed laws designed to support investment and domestic production in industries considered strategic, some of which China dominates globally. Among other goals, these policies have been adopted to (1) "de-risk" U.S. supply chains by encouraging U.S. businesses to diversify sources of supply away from China, (2) encourage Chinese policymakers to take less aggressive stances on foreign and security policy issues on which the two countries differ, and (3) further restrict China's access to technologies that would enhance the capabilities of its military.

Several years have passed since the adoption of the first of these more-restrictive measures in 2018.[1] In this report, we review the results of those policies and propose recommendations. We examine policies that address the United States' dependence on imports from China, support investment and production in domestic industries deemed critical for U.S. national security and technological leadership, and prevent U.S. technologies from being transferred to China through curbs on investment, prohibitions on transfers of technology, and other measures. Although adjacent, noneconomic topics related to recent U.S.-China relations, such as the race to dominate frontier technologies, are compelling and interesting, we did not consider them within the bounds of this study. We also limited our study to the years 2017 to 2024 because, although U.S.-China trade tensions have waxed and waned for decades, they have remained persistently high since 2017.

We categorize U.S. policy goals into efforts to promote fairer trade and to defend U.S. economic-related interests. We define *fair trade* as U.S. firms obtaining market access equal to host country firms and foreign nations fully complying with the trade agreements that they have signed with the United States. Fair trade also means that countries do not engage in uncompetitive practices, such as dumping or providing major state subsidies. We consider the defense of U.S. economic-related interests in terms of the expansion of exports and foreign trade (and investment more generally), the protection of the intellectual property of U.S. businesses, and the reduction of risks pertaining to single sources of supply.

We found that U.S. economic policies achieved limited progress in promoting fair trade but a higher degree of success in defending U.S. economic-related interests. Increases in U.S. tariffs have succeeded in reducing imports from and curbing the bilateral trade deficit with China, developments that both the Trump and Biden administrations view as resulting in fairer trade. However, U.S. policies have made little progress in ensuring fair treatment for U.S. firms in China and even less in persuading the Chinese government to reduce its subsidies and other uncompetitive state assistance to its own manufacturers, especially exporters.

[1] Office of the U.S. Trade Representative, "President Trump Announces Strong Actions to Address China's Unfair Trade," Executive Office of the President, March 22, 2018.

The United States has experienced a higher degree of success in diversifying some supply chains away from China and constraining Chinese efforts to secure sensitive technologies that could be used for commercial or military purposes. Some of these economic policies, most notably tariff increases, have come at a price, such as reduced U.S. economic growth and losses in U.S. manufacturing jobs, output, and exports.

Recommendations

Using our findings, we have drawn up the following recommendations to modify U.S. economic policies vis-à-vis China. These recommendations aim to enhance the likelihood that U.S. policies will induce the Chinese government to enact policy changes that make trade fairer and improve the United States' ability to defend its economic interests. We group our recommendations under policies related to trade, controls on technology, economic diplomacy, foreign investment, industry, and diversification of supply chains away from China.

Trade

- The U.S. government should maintain higher tariffs on imports of goods from China (1) of which China is the dominant supplier and that the Departments of Defense and Commerce consider key technologies and (2) that could undermine U.S. industries considered critical to U.S. economic or national security.
- To maintain the overall competitiveness of U.S. manufacturing and to benefit U.S. consumers, the U.S. Trade Representative should offer to negotiate reductions of U.S. tariffs on nonsensitive imports of consumer goods and manufacturing inputs from China in exchange for reductions in Chinese tariffs on U.S. goods.
- The U.S. Trade Representative should enter negotiations to join the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP). Membership in the CPTPP would better achieve U.S. economic policy goals than the Indo-Pacific Economic Framework for Prosperity. It would substantially reduce tariffs faced by U.S. exporters, eliminating the advantages that U.S. trade competitors, such as Japan and South Korea, enjoy because of duty-free access to the other members of the CPTPP.

Controls on Technology

- Congress should increase funding for the Bureau of Industry and Security and the Committee on Foreign Investment in the United States to ensure they have adequate capacity to review proposed items for export controls and proposed investments, respectively.
- Because exchanges involving government research laboratories and other government-to-government exchanges have been so one-sided, the U.S. government should continue

to decline official exchanges between U.S. government agencies and their Chinese counterparts involving applied technologies.
- The White House should organize a formal committee to coordinate economic policy toward China across U.S. government agencies.
- The U.S. Department of State should ensure that U.S. embassies, especially those located in allied and partner countries, work closely with the China House to coordinate with allies and partners when implementing new export controls or other economic policies vis-à-vis China.
- The U.S. government should continue to permit academic collaboration and exchanges with China involving basic research. It should instruct the appropriate officials in the State Department and the Department of Homeland Security to ensure that Chinese academics are able to obtain U.S. visas quickly and easily and, once a visa is granted, are able to enter the United States without interference.

Economic Diplomacy

- China prefers secretive bilateral state-dominated trade agreements with lower income countries. The State Department should instruct U.S. embassies in countries contemplating entering into these types of agreements to highlight the costs of China's approach to their foreign commercial and diplomatic counterparts.
- The State Department should work with allies to offer technical advice to countries contemplating taking on loans and investments from China to ensure that the proposed project is financially viable, does not increase the country's overall debt burden, is environmentally sound, and primarily employs local rather than Chinese workers.

Foreign Investment

- In meetings with the Chinese government, companies, and investors, U.S. government officials should make it clear that Chinese foreign direct investments that do not threaten U.S. national security continue to be welcome in the United States.
- The United States should continue to press China to treat U.S. and other foreign companies operating in China the same way it treats domestic Chinese companies.

Industry

- Government subsidies for investments and operating costs of favored industries have frequently led to overcapacity and inefficiency. To forestall these outcomes, the U.S. government should confine future subsidies to industries that have clear implications for U.S. national security and for which China is the predominant supplier. The U.S. government should cease providing subsidies to companies in sectors that are not vital

for U.S. national security and in which Chinese firms have a clear comparative advantage, such as solar cells.
- The National Science Foundation and the Departments of Energy and Defense should increase funding on research and development regarding alternatives to critical minerals, technologies that lower consumption of these minerals, and recycling technologies for these minerals.

Diversification of Supply Chains

- The Department of Commerce should set up an office responsible for strategic planning for critical materials and products important for national security. The proposed office would coordinate policy for the entire supply chain, identifying choke points where investments are needed to ensure that the United States is not reliant on China.
- The Departments of State and Commerce should work through the Minerals Supply Partnership to coordinate programs with partner countries to develop alternative sources of supply for critical materials.

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CHAPTER 1

Introduction

After China began its economic reforms in 1979, the United States pursued policies designed to bring China into international institutions governing trade and foreign investment. In the following decades, U.S.-China trade relations expanded rapidly. By the 2000s, China became the largest source of U.S. imports and one of the United States' top four trading partners, along with Canada, the European Union (EU), and Mexico. The surge in trade and investment often contributed to contentious economic relations. In the 1990s and early 2000s, the United States imposed tariffs on Chinese subsidized exports that supplanted U.S. production, criticized Beijing for placing restrictions on the activities of U.S. companies that wished to invest in China, and denounced China's theft of U.S. intellectual property. China, for its part, complained about U.S. controls on exports of technologies, constraints on imports from China, and efforts to curb purchases of Chinese products and technologies on the part of U.S. friends and allies.[1]

U.S. Presidents Donald Trump and Joe Biden have adopted considerably more-restrictive policies concerning trade, investment, and technology flows between the two countries than their predecessors. Regarding China as a formidable economic competitor, the United States has passed laws, such as the Inflation Reduction Act,[2] designed to support research and development (R&D) and increase investment and domestic production in key industries, including some in which China is dominant or threatens to become so. Increases in tariffs on Chinese goods have been adopted to "de-risk" U.S. supply chains by encouraging U.S. businesses to diversify their sources of supply away from China.[3] Other policies have aimed to curtail Chinese efforts to coerce neighboring countries through economic punishments. Still others have sought to restrict China's access to technologies that could enhance the capabilities of its military.[4]

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1 Karen M. Sutter, *U.S.-China Trade Relations*, Congressional Research Service, IF11284, September 27, 2023.

2 Public Law 117-169, Inflation Reduction Act of 2022, August 16, 2022.

3 White House, "President Biden Takes Action to Protect American Workers and Businesses from China's Unfair Trade Practices," May 14, 2024a.

4 Bureau of Industry and Security, "Commerce Implements New Export Controls on Advanced Computing and Semiconductor Manufacturing Items to the People's Republic of China (PRC)," press release, U.S. Department of Commerce, October 7, 2022a.

Several years have passed since March 2018, when the Trump administration sharply raised tariffs on Chinese imports and adopted other more-restrictive economic policy measures.[5] Given the importance of trade and investment to the livelihood of Americans, understanding the effectiveness of these measures is important. With this study, we sought to answer the following questions:

- Since 2017, how effective have U.S. economic policies toward China been?
- What are the costs of these policies to the United States?
- How can the U.S. government modify these policies to make them more effective?

Background: Developments in U.S.-China Economic Relations

The normalization of U.S.-China relations in 1979 inaugurated a relaxation of Cold War tensions and the beginning of robust economic relations. Under President Bill Clinton, the United States regarded China as an important trade partner. Despite persistent differences over such issues as Taiwan, human rights, and Tibet, the U.S. and Chinese governments emphasized the benefits of trade and maintained relatively stable relations. Clinton's successor, President George W. Bush, also emphasized trade, helping China accede to the World Trade Organization (WTO) in December 2001.[6]

Over time, China's increasing economic and military power heightened concerns in Washington. President Barack Obama oversaw a transition in U.S. policy in which more-adversarial views of China competed with a policy emphasis on economic cooperation. For example, Obama sometimes called China an "adversary" even as his first Secretary of State, Hillary Clinton, emphasized cooperation.[7]

After taking office in January 2017, Trump completed the transition to a more adversarial U.S. policy toward China. The 2017 National Security Strategy described China as "challeng[ing] American power, influence, and interests" and accused China of seeking to "erode American security and prosperity" and "make economies less free and less fair."[8] Among its top goals, the strategy called for the "promot[ion of] American prosperity."[9] In his announcement of the 2017 National Security Strategy, Trump asserted that "economic secu-

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5 Office of the U.S. Trade Representative, "President Trump Announces Strong Actions to Address China's Unfair Trade," Executive Office of the President, March 22, 2018.

6 Ramon H. Myers, Michel C. Oksenberg, and David Shambaugh, eds., *Making China Policy: Lessons from the Bush and Clinton Administrations*, Rowman & Littlefield, 2001.

7 Josh Rogin, "Obama Contradicts Clinton, Calls China an 'Adversary,'" *Foreign Policy*, October 22, 2012.

8 White House, *National Security Strategy of the United States of America*, December 2017, p. 2.

9 White House, 2017, p. 4.

rity is national security" and listed "economic vitality, growth, and prosperity [as] absolutely necessary for [U.S.] power and influence."[10]

The Trump administration published its *United States Strategic Approach to the People's Republic of China* report in early 2020, providing more details about its policy. The document described two objectives of the new "competitive" approach. The first aimed to improve the resilience of U.S. institutions, alliances, and partnerships, and the second sought to compel China to cease or reduce actions harmful to the interests of the United States and its allies and partners. The document criticized many of China's economic practices, including subsidies, a "mercantilist approach to trade and investment," and forced technology transfers.[11]

An increasing concern about U.S. manufacturing spurred a reconsideration of U.S trade policies—even though U.S. industrial output hit an all-time high in 2018.[12] The Trump administration withdrew from numerous international accords, which it judged inadequately served U.S. interests, including the Trans-Pacific Partnership (TPP). Trump also reviewed and renegotiated various trade agreements, including the North American Free Trade Agreement (NAFTA), which was replaced by the U.S.-Mexico-Canada Agreement, and the U.S.-Korea Free Trade Agreement.[13]

As part of his effort to revise trade agreements, Trump sought to renegotiate trade with Beijing. One of his major policy goals was to reduce the large bilateral trade deficit. He also sought to increase employment in U.S. manufacturing and compel the Chinese government to stop unfair practices, such as heavily subsidizing exporters and investments in favored industries that resulted in global overcapacity. After several rounds of talks with Chinese officials proved fruitless, the Trump administration increased U.S. tariffs on Chinese goods in March 2018, citing unfair Chinese trade practices and the theft of U.S. intellectual property.[14] Beijing retaliated by raising tariffs on U.S. products. Trump followed the first tariff increases with a series of measures to dramatically increase tariffs on a wide variety of additional Chinese goods.[15]

In addition to the trade war, the United States increased restrictions on Chinese investment in the United States, especially in sectors featuring technology that could threaten U.S. national security. The Committee on Foreign Investment in the United States (CFIUS)—originally formed in 1975 to regulate the flow of investments from members of the Organiza-

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[10] Jim Garamone, "Trump Announces New Whole-of-Government National Security Strategy," U.S. Department of Defense, December 18, 2017.

[11] White House, *United States Strategic Approach to the People's Republic of China*, May 20, 2020.

[12] Federal Reserve Economic Data, "Industrial Production: Total Index," interactive graph, Federal Reserve Bank of St. Louis, last updated July 17, 2024b.

[13] Paul Wiseman, "Trump Trade Policies: 4 Years of High Drama. Little Results," Associated Press, October 27, 2020.

[14] Office of the U.S. Trade Representative, 2018.

[15] Andres B. Schwarzenberg and Keigh E. Hammond, *U.S.-China Tariff Actions by the Numbers*, Congressional Research Service, R45949, October 9, 2019.

tion of the Petroleum Exporting Countries (OPEC) in the United States—is an interagency committee that oversees the national security implications of any foreign direct investment (FDI). Through its review process, CFIUS has prevented Chinese firms from acquiring U.S. companies that might have implications for U.S. national security.[16] In August 2018, Chinese efforts to acquire U.S. companies, especially in high-technology sectors, faced further hurdles with the passage of the Foreign Investment Risk Review Modernization Act (FIRRMA), which extended CFIUS's authority to review proposed and consummated foreign companies' investments in critical and emerging technologies. The U.S. government also prohibited investments by U.S. companies and individuals in Chinese companies affiliated with the People's Liberation Army. Trump strengthened the prohibitions in the Fiscal Year 2021 National Defense Authorization Act and related executive orders.[17]

Under Biden, the U.S. government has continued to view China as a great power rival and potent economic threat. The 2022 National Security Strategy characterized China as a "competitor" that had "both the intent to reshape the international order and . . . the economic, diplomatic, military, and technological power to advance that objective." The National Security Strategy criticized "autocratic governments," including China, for "abus[ing] the global economic order." It included a section that discussed measures to "advanc[e] the interests of American workers," which cited such goals as revitalizing manufacturing, ensuring fair treatment for U.S. companies abroad, and shaping international trade rules and norms to ensure fair trade.[18]

In a 2022 speech, Secretary of State Antony Blinken elaborated on the Biden administration's China policy. Blinken criticized Beijing for "undermining" international laws, agreements, principles, and institutions. He characterized the U.S. strategy in three words: "invest, align, compete." He explained that *invest* referred to strengthening the "foundations" of U.S. national power, such as the United States' democratic governance and its ability to innovate technologically. He explained that *alignment* referred to actions in common purpose with U.S. allies and partners. And he explained that *compete* referred to actions to "defend our interests" from harmful acts by China. In an example of Biden's China policy, Blinken noted that the U.S. government had launched the Indo-Pacific Economic Framework for Prosperity (IPEF), a set of regional economic agreements that address such issues as the digital economy, supply chains, clean energy, infrastructure, fair treatment for U.S. companies abroad, and corruption. He added that other relevant policies include measures to strengthen export controls, protect academic research, bolster cyber defenses, and reduce risks to U.S. supply

[16] Zhang Qingyan, Steven Croley, Xu Hui, and Peng Yi, "CFIUS and Chinese Investments in the United States—A Closed Door?" trans. by David Blumental, Steven Croley, and Xu Hui, Latham & Watkins, article reprint (No. 2352) from *Financial Times*, June 4, 2018.

[17] Humeyra Pamuk, Alexandra Alper, and Idrees Ali, "Trump Bans U.S. Investments in Companies Linked to Chinese Military," Reuters, November 13, 2020.

[18] White House, *National Security Strategy*, October 2022c.

chains.[19] On November 16, 2023, the United States and its partners signed three of the four IPEF-related agreements.[20]

Treasury Secretary Janet Yellen defined the U.S. economic approach to China in terms of ensuring the national security interests of the United States and its allies and promoting human rights, fair competition with China, and cooperation on global challenges, such as climate change. Yellen explained that the first category involves using economic levers to protect U.S. interests and values. For the second category, Yellen stated that the United States intends to "partner with [its] allies to respond to China's unfair economic practices" while continuing to invest in its own economy and advancing the United States' "vision for an open, fair, and rules-based global economic order."[21] In pursuit of these goals, in 2022, the State Department established an Office of China Coordination to coordinate information-sharing and policy with U.S. allies and partners on issues related to China.[22]

In sum, U.S. economic policies toward China have become much more antagonistic since 2017. As China's economic power has grown and its interests have diverged from those of the United States, Washington has shifted its emphasis from cooperation to competition. Under Trump and Biden, the U.S. government designated China as a competitor and emphasized economics as a critical domain of competition. U.S. policy statements make clear that U.S. policy toward China also involves building up key U.S. and allied industries in which China currently or prospectively holds dominant positions, strengthening international rules and norms favored by Washington, and persuading China to halt practices that harm U.S. economic sectors.

Goals of U.S. Economic Measures Regarding China

Our evaluation of the effectiveness of U.S. economic policies from 2017 to 2024 regarding China begins with a review of the specific goals that the policies aim to achieve. The Biden and Trump administrations share the view that the United States must compete economically with China. Employing the rhetoric of "America First," Trump emphasized fair deals and pledged to defend U.S. interests. The Biden administration added an emphasis on coordinat-

[19] Antony J. Blinken, "The Administration's Approach to the People's Republic of China," speech delivered at George Washington University, U.S. Department of State, May 26, 2022.

[20] Those agreements were the IPEF Supply Chain Agreement, the Clean Economy Agreement, and the Fair Economy Agreement (White House, "In San Francisco, President Biden and 13 Partners Announce Key Outcomes to Fuel Inclusive, Sustainable Growth as Part of the Indo-Pacific Economic Framework for Prosperity," fact sheet, November 16, 2023b). Negotiations on the trade agreement of the IPEF continue.

[21] Janet L. Yellen, "Remarks by Secretary of the Treasury Janet L. Yellen on the U.S.-China Economic Relationship at John Hopkins School of Advanced International Studies," U.S. Department of the Treasury, April 20, 2023.

[22] Nahal Toosi and Phelim Kine, "Biden Launches 'China House' to Counter Beijing's Growing Clout," *Politico*, December 16, 2022.

ing with allies and partners but otherwise maintained the focus on fairer trade agreements and the defense of U.S. interests. The speeches by Secretaries Blinken and Yellen have emphasized the overall goal of achieving fair competition with China so that each side benefits. Per the explanations offered by both senior officials, this involves measures to defend U.S. interests and enhance fair economic competition that brings benefits to both sides, which we shorten to *promote fair trade*. We will use these two categories ("defend U.S. interests" and "promote fair trade") as our main categories for evaluating the effectiveness of U.S. economic policies toward China (Table 1.1).

We have adopted the definition of *fair trade* by the U.S. Department of Commerce's International Trade Administration for this report. Per the International Trade Administration, fair trade occurs when U.S. firms and workers obtain market access equal to host country firms and when foreign nations fully comply with the trade agreements that they have signed with the United States. Fair trade occurs when countries do not engage in uncompetitive practices, such as dumping or providing major state subsidies.[23] For the broader goal of promoting fair trade, we add the following sub-objectives:

- Ensure the same access in China for U.S. products as the United States provides Chinese products.
- Reduce imports from China and the bilateral trade deficit.
- Shape international trade rules and norms.

TABLE 1.1

Goals and Sub-Objectives of U.S. Economic Policies vis-à-vis China

Goal	Sub-Objectives	Sample Policies
Promote fair trade	• Reduce bilateral trade deficit • Shape international trade rules and norms • Build up U.S. manufacturing, especially in sectors in which China dominates • Secure fair treatment for U.S. companies in China • Reduce unfair Chinese trade practices	• Tariffs • Efforts to make trade rules and norms more equitable • Investment subsidies
Defend U.S. interests	• Reduce excessive dependence on Chinese suppliers • Control transfer of key technologies	• Tariffs • Investment controls • Controls on semiconductor exports • Controls on exports of critical technologies • Industrial policies • Export licensing requirements • Addition of specific Chinese companies to the Entity List

SOURCES: Contains information from Blinken, 2022, and Yellen, 2023.

[23] International Trade Administration, "Ensuring Fair Trade," webpage, U.S. Department of Commerce, undated.

- Secure the same treatment for U.S. and other foreign companies in China as Chinese companies receive.
- Reduce unfair Chinese trade practices, especially regarding subsidies and restrictions on access by U.S. companies to Chinese markets.

These sub-objectives are evident in the economic policy goals outlined by the Trump and Biden administrations. One of the policy goals laid out in Yellen's 2023 speech was ensuring the same access in China for U.S. products as the United States provides Chinese products. Yellen stated that China "imposes numerous barriers to market access for American firms that do not exist for Chinese businesses in the United States" and called on China to end such restrictions to ensure "healthy economic competition."[24] Reducing the bilateral trade deficit was listed as a policy goal in statements by Trump; similarly, the Biden administration hailed the reduction of the bilateral trade deficit as progress toward the goal of fair trade practices.[25] Secretary Blinken's 2022 speech referenced the policy goal of shaping international trade rules and norms when he called for "reform[ing] the rules-based international order" to "make sure that it represents the interests, the values, and the hopes of all nations."[26] The sub-objective of securing the same treatment for U.S. and other foreign companies in China as Chinese companies receive was also listed in the speech by Secretary Yellen when she criticized China's mistreatment of U.S. and foreign companies and called for fair treatment to enable "healthy competition." Finally, reducing unfair Chinese trade practices was a sub-objective also listed in the Yellen speech, in which she criticized state subsidies and other unfair trade practices. She stated that the U.S. government will "take coordinated actions with our allies and partners" to counter "China's unfair economic practices."[27]

To achieve the goal of promoting fair trade, the U.S. government has not only raised tariffs on goods imported from China but also adopted industrial policies, such as the Creating Helpful Incentives to Produce Semiconductors (CHIPS) and Science Act in 2022, to support domestic R&D and manufacturing of advanced semiconductors in the United States. The Inflation Reduction Act (IRA) of 2022 includes provisions to invest in the domestic clean energy sector. The Bipartisan Infrastructure Bill also provides subsidies to companies to manufacture green technologies. Although these measures also have domestic political and other rationales, they are simultaneously seen as a means of leveling the playing field against China's substantial subsidized investments in its semiconductor and clean energy industries.

Other relevant policies include efforts to shape the terms of international trade in a manner that defines fair trade practices, set internationally agreed labor and environmental standards for traded goods, set internationally agreed rules for digital trade, and protect

[24] Yellen, 2023.

[25] White House, 2024a; Keith Bradsher, "China Is Set to Take a Hard Line on Trump's Trade Demands," *New York Times*, April 30, 2018.

[26] Blinken, 2022.

[27] Yellen, 2023.

intellectual property. Obama advanced the TPP with similar goals in mind. But after Trump withdrew, Biden subsequently put forward the IPEF, which sought more-modest goals and omitted some of the TPP's more-contentious provisions for market access to the United States. Although the U.S. government has signed the Supply Chain Agreement, the Clean Economy Agreement, and the Fair Economy Agreement,[28] it had not signed the agreement related to trade (in the IPEF) as of April 2024.[29]

Under the category of "defend U.S. interests," we add the following sub-objectives: (1) diversify supplies of critical materials and goods to reduce U.S. dependence on China and (2) control the export of sensitive technologies to China that could be used for military or commercial purposes. The first sub-objective was taken from Yellen's speech, which highlighted a "strategy" of "'friendshoring'" (i.e., developing alternative sources of supply for critical materials) and "creating redundancies in our critical supply chains with [a] large number of trading partners" to mitigate "vulnerabilities that can lead to supply disruptions."[30] The second is derived from both the Yellen and Blinken speeches, which emphasized this sub-objective as part of the broader goal of defending U.S. interests.[31]

Export controls and restrictions on Chinese direct investment in the United States are a central component of defending U.S. national and economic security. Additionally, the U.S. government has employed a wide variety of other economic tools to achieve its policy goals regarding China. These tools have included extensive use of tariffs; export controls aimed at specific entities, such as Huawei, and products, such as semiconductors; law enforcement measures to prevent the transfer of intellectual property; and other measures.[32]

Approach and Methodology

This research was undertaken to provide the U.S. government with recommendations on how to better devise and implement trade, investment, and industrial policies regarding China. To bound the study, we limited our analysis to the economic dimension. This dimension primarily includes policies to address the United States' dependence on imports from China and reduce the bilateral trade deficit, to support investment and production in domestic industries deemed critical for U.S. national security and technological leadership, and to protect U.S. technologies from being transferred to China through curbs on Chinese investment in the United States, exports of technology, and other measures. Although there are compelling

[28] White House, 2023b.

[29] Demetri Sevastopulo and Alex Rogers, "Joe Biden Halts Plan for Indo-Pacific Trade Deal After Opposition from Democrats," *Financial Times*, November 14, 2023.

[30] Yellen, 2023.

[31] Yellen, 2023; Blinken, 2022.

[32] Elizabeth Rosenberg, Peter Harrell, and Ashley Feng, *A New Arsenal for Competition: Coercive Economic Measures in the U.S.-China Relationship*, Center for a New American Security, April 24, 2020.

and interesting adjacent noneconomic topics, such as the race to dominate frontier technologies, we did not consider these topics within the bounds of this study. We also bound the study to the time frame of 2017 to early 2024 because, although they have waxed and waned for decades, U.S.-China trade tensions have remained persistently high since 2017.

We mainly relied on primary sources of quantitative economic data on U.S.-China economic activity and on qualitative information gathered from a variety of studies, news reports, and policy reports. We also assess the role of other countries, especially friends and allies, with regard to the likely success of achieving U.S. economic policy goals vis-à-vis China. Other countries play an especially important role in enacting measures to constrain China's efforts to dominate markets for critical goods and materials. We also reviewed Chinese-language policy documents and commentary to understand Chinese assessments and perspectives on China's economic relationship with the United States.

Report Outline

In Chapter 2, we focus on measures associated with the overarching goal of promoting fair trade. We analyze outcome metrics, such as changes in bilateral flows of exports and imports. We consider the effect of Chinese retaliatory measures and the ways that U.S. economic measures have affected living standards and job opportunities for Americans.

In Chapter 3, we examine measures related to the goal of defending U.S. interests. Although there is some overlap with the former category, we focus primarily on the effects of restrictions on Chinese investment in the U.S. technology sector. For a more thorough analysis, we study the effects of U.S. restrictions on five sectors involving critical minerals or key technologies. We also evaluate U.S. efforts to encourage diversification of suppliers to reduce U.S. dependence on China.

In Chapter 4, we review China's response. To provide context, we first examine China's economic situation and development goals. We then look at the short- and longer-term policies articulated by Beijing in response to recent U.S. economic policies.

In the final chapter, we provide an overall assessment of the effects of these policies on China and conclude with some recommendations as to how U.S. policies might be adjusted to be more effective. We base our suggestions on what has appeared to work and what has been less successful, drawing on the economic analyses and the assessment of changes in Chinese policies.

CHAPTER 2

Effectiveness of U.S. Policies to Achieve Fairer Trade with China

In this chapter, we assess the effectiveness of U.S. measures to bolster "fair trade" with China. This goal was implied in the Trump administration's policy statements and official speeches and explicitly identified as a key goal of the Biden administration's economic policies regarding China. We examine the effectiveness of U.S. policies to

- reduce unfair competition from Chinese imports, overall U.S. imports from China, and the bilateral trade deficit
- shape international trade rules and norms
- invest in U.S. manufacturing, especially sectors in which China dominates
- secure fair treatment for U.S. (and other foreign) companies in China
- curtail unfair Chinese trade practices.

Tariffs

As noted in Chapter 1, the U.S. government raised tariff rates in 2018 on a wide variety of goods, many of which specifically targeted China. Average U.S. tariffs on imports from China rose from 2.6 percent prior to the first tariff increase to 17.5 percent on September 1, 2019, the last time tariffs were raised. Table 2.1 shows the dates and magnitudes of these increases and the value of the imports the tariffs affected. The table also shows the increases in tariffs that China imposed in response to the actions by the United States. China increased its average tariff rate on its imports from the United States from 6.2 percent before the United States began to raise tariff rates to 16.4 percent in September 2019.[1]

Changes in Trade Flows Following the Increases in Tariffs

Figure 2.1 shows data from the U.S. Census Bureau on trade between China and the United States. The figure shows that, by 2023, U.S. imports from China had fallen to $427.2 billion,

[1] Eddy Bekkers and Sofia Schroeter, *An Economic Analysis of the US-China Trade Conflict*, Staff Working Paper ERSD-2020-04, Economic Research and Statistics Division, World Trade Organization, March 19, 2020.

TABLE 2.1
Timeline of U.S. and Chinese Tariffs

	United States		China	
Date	Average Tariff Rate (%)	Value of Imports from Preceding Year Affected by the Increased Tariffs ($ billions)	Average Tariff Rate (%)	Value of Imports from Preceding Year Affected by the Increased Tariffs ($ billions)
January 1, 2018	2.6		6.2	
March 23, 2018		3.60		
April 2, 2018	2.7		6.6	2.97
July 6, 2018	4.3	33.40	11.1	42.52
August 23, 2018		14.31		14.11
September 24, 2018	8.8	198.87	13.8	53.39
June 1, 2019	14.5		15.3	52.85
September 1, 2019	17.5	130.15	16.4	28.67
December 15, 2019	16.0	161.88	16.4	44.80
Cumulative total		487.35		113.59

SOURCE: Features data from Bekkers and Schroeter, 2020.

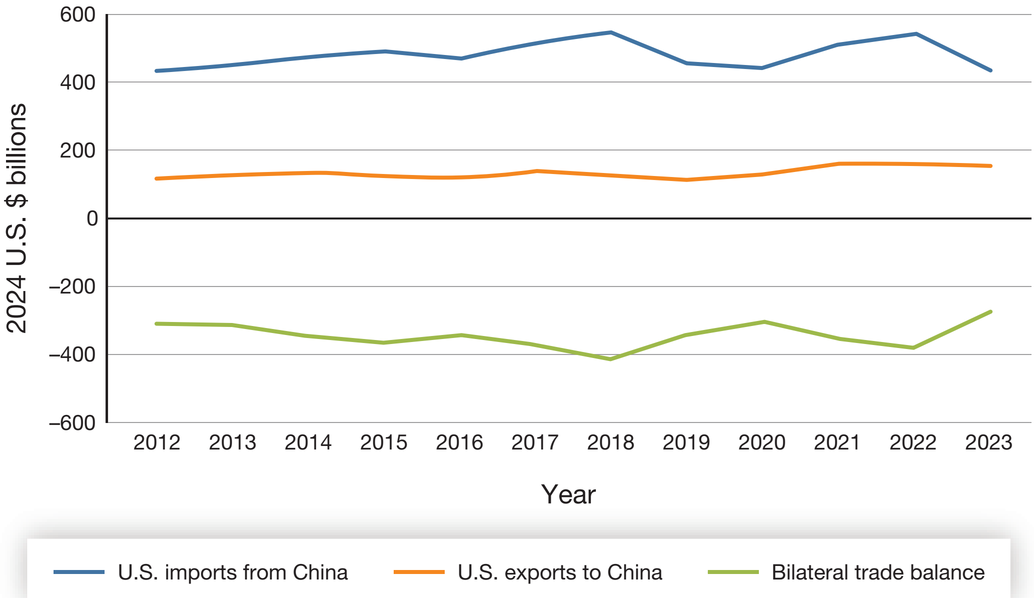
NOTE: Adding the totals for both the United States and China exceeds the reported cumulative total for two reasons. One, the base years for the row entries differ: For some entries, the previous year is 2017, for others 2018. Two, there is some overlap in terms of the volume of affected imports because the same categories of imports were targeted for different interventions.

a decline of 15.4 percent compared with 2017. Figure 2.1 also shows that U.S. imports from China in 2023 were at their lowest level since 2012. The decline in imports from China was especially sharp in 2023, when they fell 20.3 percent compared with 2022. U.S. imports from China first fell in 2019, the year following the increases in U.S. tariffs on Chinese goods. However, during the coronavirus disease 2019 (COVID-19) pandemic in 2021 and 2022, U.S. consumers sharply increased expenditures on imported goods, including from China.

In addition to the decline in the value of U.S. imports from China, China's share in total U.S. imports has fallen. According to U.S. trade statistics, China's share of total U.S. imports peaked in 2017 at 21.6 percent but fell to 13.9 percent in 2023, a decline of 7.7 percentage points. Most of this decline has been captured by Mexico and East and Southeast Asia. The share of U.S. imports from Cambodia, Singapore, South Korea, Taiwan, Thailand, and Vietnam rose 4.7 percentage points between 2017 and 2023, and Mexico's share rose 2.1 percentage points for a total of 6.8 percentage points compared with the decline in China's share of total U.S. imports of 7.7 percentage points.[2] Some of the increases in U.S. market share by these countries reflect the transfer of some Chinese manufacturing operations to those

[2] U.S. Census Bureau, "Trade in Goods with China," dataset, undated-a.

FIGURE 2.1
U.S.-China Bilateral Trade in Goods 2012–2023



SOURCE: Data are from U.S. Census Bureau, undated-a.

countries to avoid tariffs.[3] In these instances, the share of Chinese inputs in the final product remains high because some companies continue to import inputs from their Chinese suppliers for assembly in plants in Mexico and in East and Southeast Asian countries. However, some of these shifts were the result of U.S. companies attempting to diversify their supply chains by shifting imports to countries other than China.[4]

Changes in U.S. Exports to China

In contrast to U.S. imports from China, U.S. exports rose between 2017 and 2023, going from $130 billion to $147.8 billion, an increase of 13.7 percent. However, total U.S. exports rose even faster over this period—by 30.5 percent—so the share of total U.S. exports going to China fell from 8.4 percent in 2017 to 7.3 percent in 2023, a decline of 1.1 percentage points.

As trade relations deteriorated, China and the United States agreed to engage in negotiations over the tariffs. In January 2020, the United States and China concluded a phase one trade agreement that required structural reforms and other changes to economic and trade

[3] Flora Haberkorn, Trang Hoang, Gordon Lewis, Carter Mix, and Dylan Moore, “Global Trade Patterns in the Wake of the 2018–2019 U.S.-China Tariff Hikes,” Board of Governors of the Federal Reserve System, April 12, 2024.

[4] Haberkorn et al., 2024.

regimes regarding intellectual property, technology transfer, and other topics. Under this agreement, China was to increase imports of goods from the United States above their 2017 level—by $63.9 billion in 2020 and $98.2 billion in 2021.[5] These targets were not achieved. Part of the reason for the failure of the agreement owed to conflicting U.S. policy objectives and unrealistic expectations.[6] Some senior administration officials were most interested in reducing imports from China. Others were more focused on increasing U.S. exports.[7] Whatever the reason, China did not increase imports from the United States to the extent promised. In 2020, U.S. exports of goods to China were $5.4 billion less than in 2017. In 2021, China imported $21 billion more than it had in 2017, falling far short of the promised $98.2 billion.

Changes in the Balance of Trade

Import competition from China between 1990 and 2007 contributed to an estimated decline of 1,530,000 manufacturing jobs in the United States, one-quarter of the aggregate decline in U.S. manufacturing employment over that period.[8] Not surprisingly, the magnitude of Chinese imports and the bilateral trade deficit are seen as politically negative indicators of the economic relationship. One of the Trump administration's goals for the tariff increases was to reduce the United States' bilateral trade deficit with China. As can be seen in Figure 2.1, the United States' bilateral trade deficit with China in 2023 had fallen to $279 billion compared with $375 billion in 2017. The bilateral trade deficit as a share of the United States' overall deficit fell from 47 percent in 2017 to 26 percent in 2023. The decline in the deficit with China took place during a period when the overall U.S. trade deficit increased significantly from $420 billion in 2017 to $823 billion in 2023.

However, in both China and the United States, employment in manufacturing has been declining because of productivity gains. China lost 21.2 million manufacturing and construction jobs between 2012 and 2022, 10 percent of total jobs in these sectors.[9] After falling by one-third between 2000 and 2010, U.S. manufacturing employment rose 7 percent between

[5] Economic and Trade Agreement Between the Government of the United States and the Government of the People's Republic of China, signed at Washington, D.C., January 15, 2020.

[6] Chad P. Bown, "Anatomy of a Flop: Why Trump's US-China Phase One Trade Deal Fell Short," Peterson Institute for International Economics, February 8, 2021.

[7] This debate has been described as between those who favor "de-risking" economic relations with China and those who favor "decoupling." For discussions of these two views, see Emily Benson and Gloria Sicilia, "A Closer Look at De-Risking," Center for Strategic and International Studies, December 20, 2023; Alex Capri, "China Decoupling Versus De-Risking: What's the Difference?" Hinrich Foundation, December 12, 2023; Damien Cave, "How 'Decoupling' from China Became 'De-Risking,'" *New York Times*, May 20, 2023; and Agathe Demarais, "What Does 'De-Risking' Actually Mean?" *Foreign Policy*, August 23, 2023.

[8] David H. Autor, David Dorn, and Gordon H. Hanson, "The China Syndrome: Local Labor Market Effects of Import Competition in the United States," *American Economic Review,* Vol. 103, No. 6, October 2013.

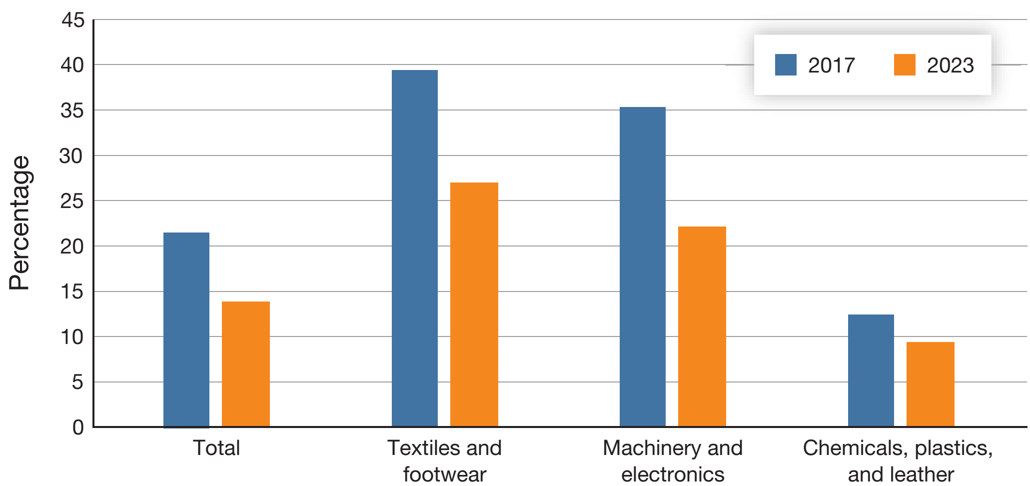
[9] National Bureau of Statistics of China, *China Statistical Yearbook 2023*, 2023, Table 4.2, "Number of Employed Persons at Year-End by Three Strata of Industry."

2012 and 2022,[10] in the same period when China's manufacturing labor force declined and the bilateral trade deficit surged.[11]

Changes in the Composition of Imports from China

The diversification of U.S. trade away from China to other countries becomes clearer by examining changes in the sources of U.S. imports by product categories. Three major product categories account for over two-thirds of U.S. imports from China: machinery and electronics, textiles and footwear, and chemicals, plastics, and leather products.[12] China's share of U.S. total imports in these three categories declined between 2017 and 2023 (see Figure 2.2). As with all categories of products imported from China, declines have been concentrated among those goods subject to increased tariffs.[13]

FIGURE 2.2
China's Share of Total U.S. Imports, by Commodity Group



SOURCES: Features information from U.S. Census Bureau, "USA Trade Online," data tool, undated-b; and Bureau of Industry and Security, "Archive: Statistical Analysis of Trade with China," statistical reports from 2014–2021, U.S. Department of Commerce, undated.

NOTE: Aggregates from the Bureau of Industry and Security and those computed from USA Trade Online differed slightly, by less than 1 percent.

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[10] Federal Reserve Economic Data, "All Employees, Manufacturing," interactive graph, Federal Reserve Bank of St. Louis, last updated July 15, 2024a.

[11] Chapter 4 provides more information on China's changing economic situation.

[12] Machinery and electronics (harmonized system [HS] categories 84 and 85), textiles and footwear (HS categories 50 through 67), and chemicals, plastics, and leather (HS categories 28 through 43).

[13] Haberkorn et al., 2024.

Competitor countries have displaced imports from China in part because of the increases in tariffs. Initially, imports from China that faced competition from other countries' imports (i.e., ones that are perfect or very close substitutes) lost more market share than products for which China was the sole provider, such as iPhones. For example, China's share of U.S. textile and footwear imports—which are labor-intensive, less sophisticated products—fell from 39 percent in 2017 to 27 percent in 2023. However, between 2017 and 2023, even China's share of U.S. machinery and electronics imports—products for which Chinese manufacturers have proprietary technologies and specialized products—fell from 36 percent to 22 percent.[14] In 2023, imports from China in every notable commodity group registered a decline in comparison with 2022.

This decline in China's share of total U.S. imports reflects transfers of some operations by non-Chinese companies out of China to reduce risks from escalating U.S.-China tensions. For example, Apple has relocated some iPhone assembly to India as part of its efforts to diversify its global production away from China.[15] Prior to 2018, 60 percent of all Chinese exports to the United States were made by factories owned by non-Chinese firms; in the computer and electronics sector, 68 percent of Chinese exports to the United States were manufactured by non-Chinese firms.[16] However, 2023 seems to have marked a turning point. The sharp decline in imports from China across so many product categories—which occurred during the five years following the imposition of higher U.S. tariffs in 2018 and 2019—suggests that U.S. importers have substantially shifted supply chains out of China to other countries. U.S. imports from China have continued to fall in 2024, although at a slower pace—2.5 percent through April 2024.[17]

Economic Costs of the Tariffs

Since 2017, several U.S. government policy goals based on increasing tariffs on Chinese imports have been achieved: The value of imports and the share of Chinese imports in total U.S. imports have fallen, as has the bilateral trade deficit. These shifts in trade with China have, however, inflicted economic costs on both China and the United States. In this section, we review estimates of those costs.

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[14] Aggregate import figures from China were taken from Bureau of Industry and Security, undated. Statistics for trade with the world and more-detailed commodity imports from China were taken from U.S. Census Bureau, undated-b. Aggregates from the Bureau of Industry and Security and those computed from USA Trade Online differed slightly, by less than 1 percent.

[15] Irene Benedicto, "Why Apple Is Manufacturing the iPhone 15 in India," *Forbes*, August 17, 2023.

[16] Mary E. Lovely and Yang Liang, "Trump Tariffs Primarily Hit Multinational Supply Chains, Harm US Technology Competitiveness," Policy Brief 18-12, Petersen Institute for International Economics, May 2018.

[17] U.S. Census Bureau, undated-b.

Costs to China

Several studies have attempted to estimate or project the cost of the 2018 tariff increases. In *World Economic Outlook, October 2019*, the International Monetary Fund (IMF) estimated a prospective cost to China of 0.3 to 0.4 percent of its 2019 gross domestic product (GDP) and sustained losses of 0.2 percent of GDP from the combined effects of U.S. tariff increases and China's retaliatory increases.[18] In current dollars, those losses would have been approximately $60 billion in 2019 and $36 billion in 2023. Economists Eddy Bekkers and Sofia Schroeter similarly estimated that the increases in Chinese and U.S. tariffs may reduce Chinese GDP by 0.2 percent per year on an ongoing basis.[19]

In a 2021 paper, Davin Chor and Bingjing Li used high-frequency data on nighttime illumination in certain Chinese localities and tied those data to measures of trade exposure at those locations based on the Chinese exporters located there. The trade exposure measure was constructed from the geocoordinates of the Chinese firms.[20] By exploiting within-grid variations over time and controlling for grid-specific contemporaneous trends, they found that each percentage point increase in exposure to higher U.S. tariffs was associated with a 0.59 percent reduction in night-time luminosity in regions where Chinese firms that are major U.S. exporters reside. They combined that finding with correlations between nighttime illumination and economic output to estimate the economic costs of the tariffs. They inferred that 2.5 percent of China's total population—those who live in the grids that bore the largest U.S. tariff shocks—experienced a 2.52 percent decrease in income per capita and a 1.62 percent decline in manufacturing employment relative to unaffected grids. They did not find significant effects from China's retaliatory tariffs on Chinese incomes or manufacturing employment.

Costs to the United States

Several studies have attempted to quantify the economic costs of the tariffs to the U.S. economy. According to the IMF estimates discussed previously, the estimated cost of the direct effects of the tariffs on the U.S. economy in 2019 was estimated at 0.18 percent of GDP; the cost in 2023 was projected to be 0.1 percent of GDP on an ongoing basis. Dollar costs would have been $39 billion and $27 billion for 2019 and 2023, respectively.[21] Bekkers and Schroeter estimated the direct cost to the U.S. economy at 0.16 percent of GDP in 2019 and projected

[18] Percentages estimated from International Monetary Fund, *World Economic Outlook, October 2019: Global Manufacturing Downturn, Rising Trade Barriers*, October 2019, Scenario Figure 1.2.1, p. 38.

[19] Bekkers and Schroeter, 2020.

[20] Davin Chor and Bingjing Li, *Illuminating the Effects of the US-China Tariff War on China's Economy*, National Bureau of Economic Research, Working Paper 29349, October 2021.

[21] Percentages estimated from International Monetary Fund, 2019, Scenario Figure 1.2.1, p. 38.

that this loss would continue. Using this estimate, the costs to the United States would have been $34 billion in 2019 and $44 billion in 2023.[22]

Mary Amiti and her coauthors found that the increases in tariffs reduced U.S. aggregate welfare by $1.4 billion per month by December 2018—$8.2 billion in total in 2018 as tariffs were repeatedly raised. They estimated the ongoing loss in U.S. welfare at $16.8 billion per year,[23] which translates to 0.08 percent of 2019 GDP, because of the deadweight losses from the tariffs on the U.S. economy.[24] The Congressional Budget Office concluded that the increases in tariffs would reduce U.S. GDP by 0.5 percent in 2020 ($107 billion) and reduce average real household income by $1,277 (in 2019 dollars) in 2020.[25]

Consistent with international trade theory and numerous studies on the economic effects of tariffs, Aaron Flaaen and Justin Pierce found that the increases in U.S. tariffs resulted in a reduction in U.S. manufacturing output, exports, and employment.[26] They estimated that U.S. manufacturers that were highly exposed to the tariffs experienced a 1.4 percent reduction in employment because of the higher costs of imported inputs and the effects of retaliatory tariffs on their exports. These losses were only partially offset by a 0.3 percent increase in manufacturing employment in the industries that the tariffs were designed to protect.[27] To illustrate the consequences of the tariffs: U.S. firms that use an input imported from China must pay the additional costs of the tariffs. This puts them at a cost disadvantage against Canadian firms that use the same input to manufacture the same product. Both sell into the North American free trade area, but the U.S. firm has to absorb the cost of the tariff on the input imported from China, while the Canadian firm does not. The declines in exports, output, and employment found by Flaaen and Pierce reflect these outcomes.

Amiti and her coauthors found that the tariffs resulted in a 1 percentage point increase in U.S. producer prices. The average rate of producer price inflation between 1990 and 2018 was just over two percentage points, so the tariffs increased the rate of producer price inflation by almost 50 percent.[28] Companies that experienced a sharp increase in tariffs on imports of inputs increased factory-gate prices by 4.1 percent.[29]

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22 Bekkers and Schroeter, 2020.

23 $16.8 billion equals $1.4 billion per month times 12 months.

24 Mary Amiti, Stephen J. Redding, and David E. Weinstein, "The Impact of the 2018 Tariffs on Prices and Welfare," *Journal of Economic Perspectives*, Vol. 33, No. 4, Fall 2019.

25 Congressional Budget Office, *The Budget and Economic Outlook: 2020 to 2030*, January 2020.

26 Aaron Flaaen and Justin Pierce, *Disentangling the Effects of the 2018-2019 Tariffs on a Globally Connected U.S. Manufacturing Sector*, Finance and Economics Discussion Series 2019-086, Divisions of Research and Statistics and Monetary Affairs, Federal Reserve Board, December 23, 2019.

27 Flaaen and Pierce, 2019.

28 Amiti, Redding, and Weinstein, 2019.

29 Flaaen and Pierce, 2019.

The economic literature on the 2018–2019 tariff increases finds that the entire cost of the tariffs has been passed through to U.S. consumers and businesses.[30] A complete pass-through of tariffs to an importing nation that is a major consumer of the products, such as the United States, is unusual. In this case, the complete pass-through is even more unusual, as Chinese exporters benefited from the depreciation of the renminbi in 2019, 2020, and 2023 compared with its rate in 2017. Studies generally find that when important import markets face abrupt increases in prices because of higher tariffs or shifts in exchange rates, exporters to the country must reduce prices to keep market share. In these instances, the cost of the tariff is shared between the importing country and the exporting country. However, there was no noticeable decrease in the price of exports from China following the tariff increases in 2018 and 2019. Lower-income groups disproportionately bore these price increases because they spend a larger share of their income on goods imported from China, such as clothing and shoes, compared with middle- and upper-income groups.[31]

In short, the increases in U.S. tariffs in 2018 resulted in reductions in U.S. manufacturing exports, output, and employment; accelerated producer and consumer price inflation; and diminished household welfare, especially for lower-income households.

Proposed Changes to International Trade Rules and Norms

Another set of policies pursued by the U.S. government to promote fair trade consists of proposed changes to international trade rules and norms. Continuing the pursuit of establishing and revising international trade rules and norms could advance several key U.S. goals regarding fair trade. Establishing new trade agreements could codify standards of fair treatment for companies in other countries, and revising trade rules could standardize expectations for fair competition by addressing such issues as government subsidies. The WTO has historically served as a venue for addressing such issues, but gridlock over disputes has raised questions about its role in resolving international trade disputes.[32] Outside the WTO, the U.S. government has also often called on China to ensure fair treatment for U.S. companies.

Since the 2010s, the U.S. government has focused on regional trade agreements as a means of countering Chinese influence and advancing fair trade goals. Although Trump successfully negotiated the United States–Mexico–Canada Agreement to replace NAFTA, he abandoned the Obama-era TPP agreement and did not propose a substitute regional trade agree-

[30] Fajgelbaum, Pablo D., and Amit K. Khandelwal, "The Economic Impacts of the US–China Trade War," *Annual Review of Economics*, Vol. 14, August 2022.

[31] Liang Bai and Sebastian Stumpner, "Estimating US Consumer Gains from Chinese Imports," *American Economic Review: Insights*, Vol. 1, No. 2, September 2019.

[32] Council of Councils, "The WTO at a Crossroads: What the Failed Ministerial Conference Means," Council on Foreign Relations, March 6, 2024.

ment for the Indo-Pacific.[33] Biden has advanced the IPEF, which omitted the TPP's provision to open U.S. markets to all signatories but included provisions to strengthen supply chain resilience and update digital trade terms. As previously mentioned, the U.S. government has signed agreements on three of the four pillars of the IPEF but has not agreed to the trade pillar because of political opposition in the U.S. Congress.[34]

Despite its failure to formally accede to the TPP, the United States has had some success in influencing international trade norms and rules. As of 2024, 12 countries had acceded to the treaty, which has subsequently been renamed the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP). The CPTPP includes provisions that address several issues of concern to the United States, including state intervention in markets and protections for intellectual property rights.[35] Thus, U.S. efforts to revise international rules and standards in a manner that promotes fair trade have achieved at least some success.

To advance its fair-trade goals and counter China's Belt and Road Initiative (BRI), the United States has introduced a series of initiatives aimed at upholding its key role in shaping global trade rules and norms. During the G7 meeting in 2021, the United States announced and spearheaded an initiative titled Build Back Better World, later rebranded as the Partnership for Global Infrastructure and Investment (PGII).[36] This initiative represents the largest Western effort thus far to counter influence garnered by China through BRI. Biden stated that, over five years, the United States would mobilize $200 billion in grants and investment from the federal and private sectors to support projects in low- to middle-income countries.[37] The PGII focuses on investments in four areas: climate change, global health, digital infrastructure, and gender equality.[38] Unlike BRI, which is unilaterally managed and supervised by the Chinese government, the PGII is a multilateral effort that relies on commitments from the United States and its G7 partners. The effectiveness of the initiative in countering BRI remains undetermined, primarily because of the uncertain nature of project finance; the inability to swiftly launch projects, compared with China, because of much longer reviews concerning the environmental and financial viability of the proposed projects; and skepticism from private sector companies regarding returns on investment.[39] These challenges

[33] Katie Lobosco, "NAFTA Is Officially Gone. Here's What Has and Hasn't Changed," CNN, July 1, 2020.

[34] Ji Siqi, "Nations in Biden's Indo-Pacific 'Framework' Are Losing Interest, Trade Group Official Warns," *South China Morning Post*, January 18, 2024.

[35] Dharshini David, "CPTPP: UK Agrees to Join Asia's Trade Club but What Is It?" BBC, July 15, 2023.

[36] White House, "Partnership for Global Infrastructure and Investment at the G7 Summit," fact sheet, June 13, 2024b.

[37] White House, "President Biden and G7 Leaders Launch Build Back Better World (B3W) Partnership," fact sheet, June 12, 2021a.

[38] White House, 2024b.

[39] Simone McCarthy, "China's Belt and Road Is Facing Challenges. But Can the US Counter It?" CNN, August 22, 2022.

have made it difficult for the United States to provide a viable alternative to BRI for lower- to middle-income countries.

Another initiative, spearheaded by the United States through the G20 and titled the India–Middle East–Europe Corridor (IMEC), has made little progress. Announced during the G20 summit in September 2023, IMEC aims to better facilitate trade among India, the Arabian Gulf states, and Europe.[40] Another strategic goal of IMEC is to counter China's increasing engagement with Saudi Arabia and the United Arab Emirates. IMEC focuses not only on trade connections but also on the development of electricity and digital infrastructure, along with pipelines for exporting clean hydrogen. IMEC remains largely in the planning stages, however.[41]

In Africa, the Lobito Corridor, known as the Trans-African Corridor, aims to link Angola, the Democratic Republic of the Congo, and Zambia, ultimately extending to the Indian Ocean. The successful implementation of the Lobito Corridor would facilitate increased production and exports of critical minerals (such as copper and cobalt), reducing dependence on China for these critical minerals.[42] Both IMEC and the Lobito Corridor represent substantial endeavors that will require many years to implement.

Alongside these global initiatives, the United States has intensified its efforts to reduce the dependence of Asian countries on China. These endeavors include the Indo-Pacific Economic Framework, the Japan-U.S. Clean Energy Partnership, the Japan-U.S.-Mekong Power Partnership, the U.S.-Taiwan Economic Prosperity and Partnership Dialogue, and the Minerals Security Partnership. These regional initiatives serve to strengthen collaboration among allies and partners. By concentrating on specific areas and sectors within the supply chain, the United States is encouraging countries to reduce their dependence on China. Many of these initiatives have just gotten underway, so it is too soon to assess their effectiveness.

The U.S. government has also demanded that China ensure fair treatment of U.S. companies. In 2024, for example, Blinken called on his Chinese counterparts to provide a "level playing field" for U.S. companies operating in China.[43] However, Chinese officials have made little effort to accommodate these demands.

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[40] White House, "World Leaders Launch a Landmark India-Middle East-Europe Economic Corridor," fact sheet, September 9, 2023a.

[41] White House, 2023a.

[42] David Sacks, "Will the U.S. Plan to Counter China's Belt and Road Initiative Work?" blog post, Council on Foreign Relations, September 14, 2023.

[43] Simon Lewis, "In China, Blinken Urges Fair Treatment of American Companies," Reuters, April 25, 2024.

Industrial Policies

Under the Biden administration, the U.S. government has passed several pieces of legislation supporting industrial policies. The Bipartisan Infrastructure Law (BIL), signed into law in November 2021 and also known as the Infrastructure Investment and Jobs Act, funds improvements to transportation infrastructure and some investments in critical materials.[44] The CHIPS and Science Act authorizes federal funding for research, development, and manufacturing of semiconductors in the United States.[45] And the IRA, passed in 2022, aims to strengthen the United States' production of green energy through investments in relevant technologies and start-up manufacturers.[46] The administration has also used the Defense Production Act (DPA) to provide grants for R&D and investments in the supply chain for critical minerals. These pieces of legislation were passed and are being used, in part, to advance fair trade by countering Chinese industrial policies, thereby creating a more level playing field for U.S. companies to compete against Chinese firms in industries deemed critical to U.S. economic and national security. Many of the investment projects have just gotten underway (as of 2024), but preliminary assessments indicate that investment in manufacturing has generated significant growth. Whether these investments will increase the number of workers in manufacturing is unclear, however, as much of the work has been automated.[47] However, U.S. government investment in infrastructure should improve the competitiveness of U.S. exporters.

Conclusion

The U.S. government under Trump and Biden has upheld fair trade as a key goal for its economic policy regarding China. The sub-objectives that make up this goal have achieved varying degrees of success (Table 2.2). Relevant actions to achieve this goal include raising tariffs, proposing changes to regional trade rules and norms, and supporting industrial policies. The most significant measure undertaken to achieve this goal has been the tariff increases adopted by the U.S. government in 2018 and 2019. These increases have achieved the goal of reducing the value of Chinese imports and narrowing the bilateral trade deficit. Other U.S. initiatives to promote fair trade with China, such as regional trade agreements, have made less progress. The U.S. government's effort to shape international trade rules and

[44] Federal Transit Administration, "Bipartisan Infrastructure Law," webpage, U.S. Department of Transportation, last updated November 15, 2023.

[45] White House, "CHIPS and Science Act Will Lower Costs, Create Jobs, Strengthen Supply Chains, and Counter China," fact sheet, August 9, 2022b.

[46] Christina DeConcini, Jennifer Rennicks, and Shannon Wood, "One Year in, How the Inflation Reduction Act Is Creating a Manufacturing Resurgence in the US," World Resources Institute, August 9, 2023.

[47] Timothy Aeppel, "US Employment Boom Leaves Factory Workers Behind," Reuters, April 4, 2024.

TABLE 2.2
Assessment of U.S. Policies to Promote Fair Trade

Sub-Objective	Assessment
Reduce imports from China	Success
Reduce bilateral trade deficit	Success
Increase U.S. exports to China	Mixed
Diversify supply chains	Mixed
Shape international trade rules and norms	Mixed
Convince China to treat U.S companies equally with Chinese companies	Failure
Convince China to stop its unfair trade practices	Failure

SOURCES: Contains information from Blinken, 2022, and Yellen, 2023.

norms has yielded some fruit, as many countries accepted proposals associated with the TPP. However, the United States decided not to accede to the treaty. U.S. efforts to compel China to change how it treats U.S. companies operating in China and curtail its use of subsidies to give its companies a competitive advantage have borne little fruit. Domestically, the U.S. government is providing very large subsidies for industries deemed critical, and investment in new plants in these industries has increased sharply. The U.S. government is also attempting to diversify sources of supply away from China for a variety of products. As of 2024, it has made some progress toward diversification, but many products deemed critical are still primarily sourced from China.

U.S. policies to advance fair trade have also imposed costs on U.S. businesses, workers, and consumers. Notably, the tariff increases resulted in a reduction in overall welfare for the United States. They contributed to reductions in U.S. manufacturing exports, output, and employment; accelerated producer and consumer price inflation; and diminished household welfare.

CHAPTER 3

Effectiveness of U.S. Economic Policies to Defend U.S. Interests

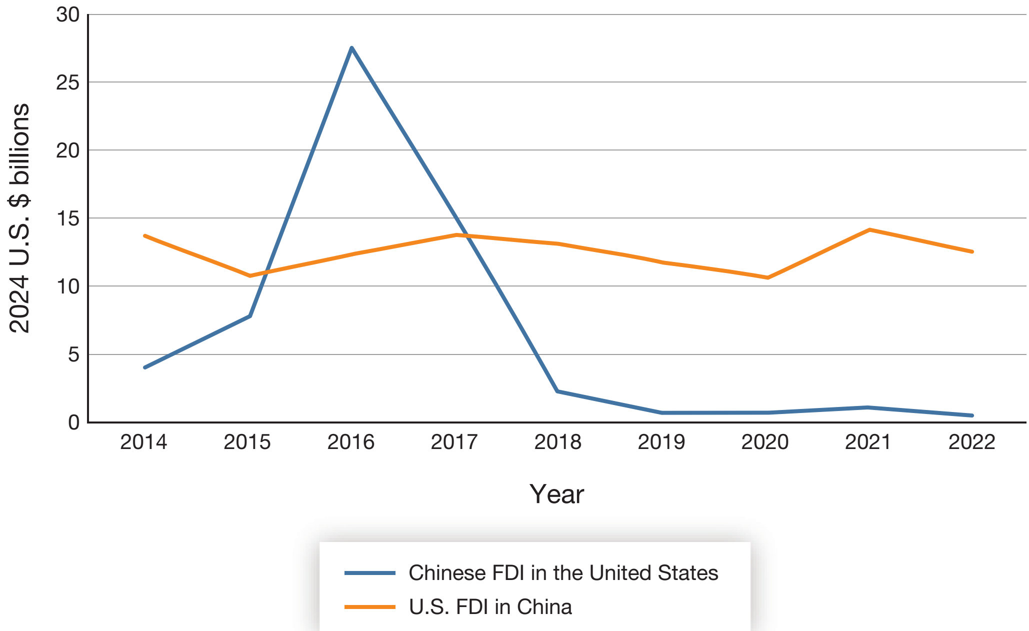
In this chapter, we evaluate the effectiveness of U.S. policies designed to protect the economic interests of the United States and its allies and partners. In particular, we focus on restrictions on the export of technologies that could grant China advantages in market sectors featuring advanced technologies, we review the effects of restrictions on Chinese investment in U.S. key industries, and we consider the effects of restrictions on U.S. investment in Chinese technology sectors that could be viewed as threats to U.S. interests. To facilitate a more thorough analysis, we conducted case studies on five product and material sectors in which the U.S. government has adopted restrictions on trade and investment: polysilicon, lithium, graphite, rare earth minerals, and advanced semiconductors.

Restrictions on China's Direct Foreign Investment in the United States

CFIUS is an interagency committee authorized to review certain transactions involving foreign investment in the United States and certain real estate transactions by foreign persons. These reviews are meant to determine the effects of such transactions on the national security of the United States.[1] Around the mid-2010s, U.S. national security policymakers and professionals became increasingly concerned that the legislation establishing CFIUS was inadequate for the U.S. government to intervene when Chinese entities were making minority investments in U.S. technology companies. Although these investments did not provide investors with controlling stakes, they could permit the transfer of important U.S. technologies to China. The U.S. government's worries stemmed from dramatic increases in Chinese investments in U.S. technology companies beginning in 2014 (Figure 3.1). Although Chinese FDI in 2014 hovered around $3.9 billion, by 2016, those investments ballooned to $27.4 billion. In 2017, Chinese FDI fell to $15 billion, much lower than 2016 but still nearly four times the 2014 investment total. During this time, Alibaba and Tencent (two Chinese technology

1 U.S. Department of the Treasury, "The Committee on Foreign Investment in the United States (CFIUS)," webpage, undated.

FIGURE 3.1
Flows of Foreign Direct Investment Between China and the United States



SOURCES: Data on Chinese FDI in the United States are from Bureau of Economic Analysis, "Direct Investment and Multinational Enterprises (MNEs)," database, U.S. Department of Commerce, last updated July 23, 2024. Data on U.S. FDI in China are from multiple editions of the National Bureau of Statistics of China, *China Statistical Yearbook*, various years (1999–2023).

companies) and Hone Capital (a Chinese venture capital fund) began to aggressively invest in U.S. technology start-ups.[2] For example, by 2017, Tencent had invested in over 40 technology firms, including Snap and Tesla.[3] The rapid increase in Chinese FDI, and especially the types of companies that Chinese investors were targeting, generated concerns in the U.S. intelligence community and in Congress about loss of critical U.S. technologies to China.[4]

The Foreign Investment Risk Review Modernization Act

These investment-related concerns about national security and U.S. leadership in emerging technologies led to the drafting and passage of FIRRMA, which became law on August 13,

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[2] Zimmerman, Evan J., "The Foreign Risk Review Modernization Act: How CFIUS Became a Tech Office," *Berkeley Technology Law Journal*, Vol. 34, No. 4, 2019.

[3] Steven Russolillo and Wayne Ma, "Tencent Continues to Snap Up Stakes in U.S. Startups," *Wall Street Journal*, November 8, 2017.

[4] Andres B. Schwarzenberg, *U.S.-China Investment Ties: Overview and Issues for Congress*, IF11283, Congressional Research Service, August 7, 2019.

2018.[5] FIRRMA strengthens and modernizes CFIUS to better address concerns about U.S. national security and U.S. technological leadership. The law also broadens the authority of CFIUS to review investments beyond mergers, acquisitions, and takeovers and potentially block minority investments, such as specified noncontrolling investments and real estate transactions involving foreign persons. Under FIRRMA, CFIUS has the authority to review investments that might pose potential risks to national security and U.S. leadership in a specific technology. FIRRMA also permits CFIUS to look at past patterns of behavior by the investor or the investor's home country when determining whether the investment is not in the U.S. national interest. Not surprisingly, legislators were highly concerned about China and inserted a reporting requirement about Chinese investment in the United States as part of the legislation.[6]

Changes in Foreign Direct Investment Flows Between China and the United States

Although China has invested massively in U.S. portfolio investments, especially purchases of U.S. government bonds, Chinese FDI in the United States was small until 2014. Beginning in 2014, however, Chinese FDI surged. This surge took place during a period when venture capital investment in information technology and other high-technology sectors was increasing rapidly in the United States. FIRRMA abruptly changed this trend. Following its passage in 2018, Chinese FDI in the United States fell sharply. By 2022, it had fallen to just $427 million, less than 2 percent of the inflow in 2016 (Figure 3.1).

In contrast, flows of U.S. FDI to China have been much less variable, fluctuating between $11 billion and $14 billion per year between 2014 and 2022, the latest year for which we have data (Figure 3.1). This relatively constant flow of funds reflects reinvestments in China by U.S. companies that have operations there, as well as new investments from the United States. Although a 2022 survey by the American Chamber of Commerce in Shanghai found that 19 percent of U.S. companies in China were cutting their investments, a large majority of respondents said they would maintain or increase their investments.[7] However, U.S. companies are much more negative about doing business in China than in the past, which is reflected in the survey.

Case Studies of Five Key Products and Materials

The enactment of FIRRMA in 2018 resulted in a dramatic decline in Chinese access to U.S. technology through FDI. In addition to the passage of FIRRMA, the U.S. government has

5 U.S. Department of the Treasury, undated.

6 Zimmerman, 2019.

7 Evelyn Cheng, "American Companies Increasingly Look Outside of China After Covid," CNBC, October 27, 2022.

passed legislation and carried out other policies to diversify U.S. supply chains and restrict Chinese access to advanced technologies. We illustrate the variety of policies and their relative effectiveness through five case studies of specific products and materials: polysilicon, lithium, graphite, rare earths, and advanced semiconductors and semiconductor manufacturing equipment. For each case study, we assess the relative success of efforts to diversify U.S. supply chains or constrain China's access to advanced technologies.

Case Study 1: Polysilicon

Polysilicon is a high-purity, crystallized version of silicon and a key material in the production of photovoltaic cells, from which solar panels are constructed. Manufactured by processing metallurgical-grade silicon (derived from quartzite) to higher levels of purity, polysilicon is then melted into ingots, which are sliced into wafers before they are processed into solar cells.[8] Polysilicon is also used in the production of semiconductors other than solar cells.

Trends in Global Market Share

Over the past decade, China has become the world's largest producer of polysilicon. During this period, global demand for polysilicon has soared because of the rise in production of solar panels. In 2012, China produced 30 percent of the global production of solar-grade polysilicon; by 2021, China's market share had reached 80 percent. In 2022, seven of the top ten polysilicon companies by production volume were Chinese (Table 3.1).[9]

Chinese Policies

In the 1990s and early 2000s, China imported most of its polysilicon from Europe and the United States.[10] However, following its 11th Five-Year Plan (2006 to 2011), the Chinese government subsidized Chinese companies engaged in the production of polysilicon and other steps in the solar panel supply chain.[11] Chinese companies also signed technology transfer agreements with European companies to learn how to process highly purified silicon.[12] As a consequence, Chinese companies ramped up production and have come to dominate the global polysilicon market.

In 2014, following an investigation by the Chinese government that found that Chinese manufacturers "suffered substantial harm" because of U.S. subsidies, China levied anti-

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[8] Edward Lees and Ulrik Fugmann, "What You Need to Know About Polysilicon and Its Role in Solar Modules," *Viewpoint*, BNP Paribas Asset Management, October 13, 2021.

[9] Kelly Pickerel, "China's Share of the World's Polysilicon Production Grows from 30% to 80% in Just One Decade," *Solar Power World,* April 27, 2022.

[10] Yasmina Abdelilah, Heymi Bahar, François Briens, Piotr Bojek, Trevor Criswell, Kazuhiro Kurumi, Jeremy Moorhouse, Grecia Rodríguez, and Kartik Veerakumar, *Special Report on Solar PV Global Supply Chains*, International Energy Agency, August 2022, p. 108.

[11] Shannon Osaka, "How 'USA-First' Failed the Solar Industry," *Grist,* May 19, 2022.

[12] Emily Feng, "How Did China Become the World's Dominant Polysilicon Producer?" *NPR,* July 6, 2021.

TABLE 3.1
Top Ten Global Polysilicon Manufacturing Companies, by Output in 2022

Company	Country	Estimated Production Capacity (metric tons)	Global Rank
Tongwei Co., Ltd.	China	345,000	1
GCL Technology Holdings Ltd.	China	300,000	2
Daqo New Energy Corp.	China	240,000	3
Xinte Energy Co., Ltd.	China	200,000	4
Xinjiang East Hope New Energy Co., Ltd.	China	130,000	7
Asia Silicon (Qinghai) Co., Ltd.	China	92,000	6
Wacker Chemie AG	Germany/United States	85,000	5
OCI Company Ltd./OCI Holdings Co., Ltd.	South Korea/Malaysia	41,500	8
Hemlock Semiconductor Operations LLC	United States	20,000	9
Shaanxi Non-Ferrous Tianhong REC Silicon Mat. Co., Ltd.	China	19,300	10

SOURCE: Estimated production capacities and global rankings are from Johannes Bernreuter, "Polysilicon Manufacturers," Bernreuter Research, June 29, 2020.

dumping duties on imports of U.S. and South Korean polysilicon—a move that was part of a broader trade conflict over the solar panel supply chain.[13] China's introduction of duties on polysilicon imports and the declining cost of polysilicon production in China (due, in large part, to the subsidies provided by the Chinese government) contributed to the loss of global market share by Hemlock Semiconductor, a U.S.-based company jointly owned by Corning Inc. and Shin-Etsu Handotai. Hemlock was the global leader in the polysilicon market from 1994 until 2011, but its global market share fell sharply after 2013.[14]

In addition to expanding the production of polysilicon, the Chinese government has supported the construction and expansion of quartzite mines. Many of the world's active quartzite mines are in China, although there are significant quartzite reserves in the United States, Brazil, and Norway.[15]

[13] "China Sets Final Duties on U.S. Solar Materials," Reuters, January 20, 2014.

[14] Heather Jordan, "Hemlock Semiconductor to Cut 100 Jobs," *MLive*, October 18, 2017.

[15] Lees and Fugmann, 2021.

U.S. Policy Response

The concentration of solar manufacturing capacity in China over the past decade raised concerns among U.S. manufacturers and U.S. officials about risks to the supply chain.[16] U.S. solar companies pushed for increased production of polysilicon outside China, arguing that "a less-concentrated supply chain will be more resilient, emit less carbon, and circumvent companies accused of using forced labor."[17]

In response to Chinese actions and the loss of global market share by U.S. companies, in 2012, the Obama administration imposed tariffs on solar panels and other solar power equipment from China. The Trump administration imposed additional tariffs on Chinese solar equipment in 2018, and the Biden administration extended those tariffs, with some exemptions.[18]

U.S. authorities also passed legislation to address another source of Chinese uncompetitive behavior. Chinese polysilicon manufacturers have long benefited from reduced labor costs by using forced labor in the Xinjiang Uyghur Autonomous Region. The Hoshine Silicon Industry factory, for example, includes two facilities that CNN has identified as detention centers for Uyghurs.[19] According to the U.S. Department of Labor, "nearly half" of global polysilicon production comes from the Xinjiang region of China.[20] The Department of Labor estimates that 100,000 Uyghurs and other ethnic minorities "may be working in conditions of forced labor following detention in re-education camps."[21]

Passed by Congress in December 2021, the Uyghur Forced Labor Prevention Act came into effect in June 2022. The law bars all goods that are fully or partially produced in the Xinjiang Uyghur Autonomous Region from entry into the United States on the grounds that they may be produced using forced labor. Polysilicon is one of just a few products designated as a "high-priority" sector for enforcement.[22] The Biden administration also

[16] Prachi Patel, "The Inflation Reduction Act vs. China's PV Dominance," *IEEE,* October 15, 2022.

[17] Matt Blois, "The US Solar Industry Has a Supply Problem," *Chemical and Engineering News*, September 18, 2022.

[18] Osaka, 2022.

[19] Michael Shellenberger, "China Helped Make Solar Power Cheap Through Subsidies, Coal and Allegedly, Forced Labor," *Forbes,* May 19, 2021.

[20] U.S. Department of Labor, "Traced to Forced Labor: Solar Supply Chains Dependent on Polysilicon from Xinjiang," infographic, undated.

[21] Bureau of International Labor Affairs, "Against Their Will: The Situation in Xinjiang," webpage, U.S. Department of Labor, undated.

[22] Public Law 117-78, An Act to Ensure That Goods Made with Forced Labor in the Xinjiang Uyghur Autonomous Region of the People's Republic of China Do Not Enter the United States Market, and for Other Purposes, December 23, 2021.

banned imports of other solar materials from Hoshine in 2021, citing allegations of using forced labor.[23]

The U.S. response to China's grip on the solar panel supply chain extends beyond tariffs and legislation about forced labor. The 2022 IRA provides a tax credit for photovoltaic modules and their subcomponents, including solar-grade polysilicon, produced in the United States.[24] Congress estimated that the tax credits for elements of clean energy production—including wind turbines, solar panels and their components, batteries for electric vehicles (EVs) and the power grid, and related mineral mining and processing—would amount to more than $30 billion in the decade following implementation. The IRA also includes tax incentives for generating electricity using clean energy sources, a measure that is spurring the installation of more solar projects. Under the IRA, the U.S. government is providing loan guarantees for investments in new factories that manufacture clean energy–related components as well.[25]

Effectiveness of U.S. Policy Measures

Neither tariffs nor the legislation on forced labor in Xinjiang appears to have substantially affected China's dominance in the polysilicon industry. However, although it is still too early to tell, the IRA could diminish China's share of polysilicon supplies by expanding the ranks of U.S.-based suppliers and U.S. production.

Analysts believe that U.S. tariffs have not yet had their intended effect of bolstering the domestic solar cell and solar panel industries by protecting U.S. manufacturers from Chinese competitors. Rather, the tariffs appear to have increased the costs of installing utility-scale solar panels in the United States.[26] Some argue that tariffs implemented before 2012 could have had a more substantial effect by deterring Chinese state subsidies but that, after that year, it was too late. "Both established U.S. manufacturers of conventional silicon solar panels and venture-funded Silicon Valley start-ups developing new solar technologies failed in the face of a flood of cheap Chinese silicon panels," one analyst wrote in 2018.[27] There is some evidence that the tariffs have increased solar panel assembly in the United States using imported cells, but the growth in U.S. assembly capacity has been mostly small-scale and has not met U.S. demand. Most U.S. solar industry groups oppose the tariffs, arguing

23 Michael Martina, Karen Freifeld, and David Shepardson, "U.S. Bans Imports of Solar Panel Material from Chinese Company," Reuters, June 24, 2021.

24 Solar Energy Technologies Office, "Federal Tax Credits for Solar Manufacturers," webpage, U.S. Department of Energy, undated.

25 Timothy C. Brightbill, Laura El-Sabaawi, and Paul A. Devamithran, "The Inflation Reduction Act Provides Potential Game-Changing Benefits for U.S. Solar Industry," Wiley, August 15, 2022.

26 Osaka, 2022.

27 Varun Sivaram, "Trump's Solar Tariffs Create Far More Losers than Winners," Council on Foreign Relations, January 23, 2018.

that "'there's now more than a decade of evidence proving that tariffs do not encourage U.S. solar manufacturing.'"[28]

The Uyghur Forced Labor Prevention Act has not had a significant effect on Chinese global sales of polysilicon. In response to the Uyghur Forced Labor Prevention Act, Chinese solar companies are reportedly dividing their supply chains, "creating a Xinjiang-free line for the U.S. while continuing to supply other global clients from factories in the region."[29] The U.S.-bound supply chain accounts for between 7 and 14 percent of the total production of Chinese solar companies. Because silicon metal from Xinjiang is used as feedstock in the production of polysilicon by Chinese companies, even Chinese companies located outside Xinjiang may use components built with forced labor.[30] The U.S.-bound supply chain sources polysilicon produced by companies based in the United States and Germany, creating a higher-priced market for non-Chinese polysilicon. The import ban does appear to have had some effect on polysilicon market share. In 2020, polysilicon produced in Xinjiang accounted for about 45 percent of global polysilicon production, but that share dropped to 35 percent in 2022.[31]

Because non-Chinese sources of polysilicon are adequate to meet the needs of the U.S. market, the law does not appear to have had a significant effect on Chinese state policies and labor practices in Xinjiang.[32] Other major importers will likely need to pass similar laws to have a significant effect on Xinjiang's share of global polysilicon markets. The EU, for example, passed legislation in April 2024 banning the import of products made with forced labor.[33]

Since its passage, the IRA has reportedly "spurred hundreds of billions of dollars in investment in clean technology."[34] Hemlock Semiconductor and Wacker Chemie AG are increasing output of polysilicon, and REC Silicon has reopened a plant in Moses Lake, Washington.[35] In contrast to Hemlock Semiconductor and Wacker, which manufacture polysilicon for semiconductors, REC Silicon is making polysilicon to be used in solar panels. Despite the launch of construction of 52 new domestic solar manufacturing facilities from August

[28] Osaka, 2022.

[29] Lili Pike, "Has the U.S. Campaign Against Uyghur Forced Labor Been Successful?" *Foreign Policy,* August 21, 2023.

[30] Kelly Pickerel, "The Global Polysilicon Market Is Entering Another Severe Oversupply Situation," *Solar Power World*, November 28, 2023.

[31] Pike, 2023.

[32] Pike, 2023.

[33] European Parliament, "Towards an EU Ban on Products Made with Forced Labour," press release, October 16, 2023.

[34] Justin Worland, "How the Inflation Reduction Act Has Reshaped the U.S.—and the World," *Time,* August 11, 2023.

[35] Ivan Penn, "Key Solar Panel Ingredient Is Made in the U.S.A. Again," *New York Times*, April 25, 2024.

2022 through July 2023 (because of funding from the IRA),[36] U.S. assemblers of solar panels still struggle, as Chinese manufacturers have moved assembly to Cambodia, Malaysia, Vietnam, and Thailand. These assembly operations have cut prices by 50 percent compared with the previous year (2022 to 2023), making U.S. assembly of solar panels uncompetitive.[37] Consequently, although the IRA has helped existing manufacturers of polysilicon for semiconductors, it will be difficult for the United States to compete with Chinese manufacturers of solar cells and solar modules, so demand for U.S. polysilicon for solar panels is likely to be limited.

Case Study 2: Lithium and Lithium-Ion Batteries

Lithium is a soft, silver-white, flammable, alkaline metal and the lightest metal in the periodic table. It is used for air purification and in metallurgy, ceramics, glass, pharmaceuticals, polymers, and batteries.[38] Lithium is considered a critical mineral because it is a key component in rechargeable lithium-ion batteries. Lithium-ion batteries are more energy-efficient and have longer life cycles than other commercial rechargeable batteries. They are used in portable consumer electronics, grid-scale energy storage, aerospace applications, and EVs, among other products. Consumption of lithium has skyrocketed in recent years as sales of EVs have taken off. Lithium-ion batteries accounted for 87 percent of global lithium consumption in 2023.[39]

Trends in Global Market Share

Australia and Chile, not China, are the largest sources of raw lithium in the world. Australia extracts lithium from granitic pegmatite and other ores from open pit mines. Lithium output from Australia accounted for 45 percent of global production in 2023 (Table 3.2). Chile, which extracts lithium from brines, is the second largest source of the material, accounting for 23 percent of global production. China ranks number three, accounting for 17 percent. Because of explosive growth in demand for lithium and the resulting rise in prices, several other countries have been increasing output. The United States, for example, has expanded production at operations in Nevada and from other sources; over a dozen companies are exploring for or investing in U.S. lithium extraction operations.

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[36] American Clean Power, *Clean Energy Investing in America*, August 2023, p. 2.

[37] Evan Halper, "U.S. Solar Companies, Imperiled by Price Collapse, Demand Protection," *Washington Post*, April 25, 2024.

[38] Dwight C. Bradley, Lisa L. Stillings, Brian W. Jaskula, LeeAnn Munk, and Andrew D. McCauley, "Lithium," Professional Paper 1802–K, in Klaus J. Schulz, John H. DeYoung, Jr., Robert R. Seal II, and Dwight C. Bradley, eds., *Critical Mineral Resources of the United States—Economic and Environmental Geology and Prospects for Future Supply*, U.S. Geological Survey, 2017.

[39] Brian W. Jaskula, "Lithium," in U.S. Geological Survey, *Mineral Commodity Summaries 2024*, U.S. Department of the Interior, 2024.

TABLE 3.2
Top Global Producers of Raw Lithium in 2022 and 2023

Major Lithium Miners	Country	Output 2022 (metric tons)	Output 2023 (metric tons)
Mineral Resources Ltd. (Australia), Pilbara Minerals (Australia), Tianqi Lithium (China)	Australia	74,700	86,000
Albemarle Corporation (U.S.), Sociedad Química y Minera (Chile)	Chile	38,000	44,000
Tianqi Lithium (China), Arcadium Lithium (U.S.), Ganfeng Lithium (China), Sichuan Yahua Industrial (China)	China	22,600	33,000
Arcadium Lithium (U.S), Ganfeng Lithium (China), Lithium Americas (Canada)	Argentina	6,590	9,600
	Brazil	2,630	4,900
Lithium Americas (Canada)	Canada	520	3,400
	Zimbabwe	1,030	3,400
Albemarle (U.S.), Arcadium Lithium (U.S), Lithium Americas (Canada)	United States	~2,900[a]	5,000[b]
	Portugal	380	380
	Total[c]	~149,000	~190,000

SOURCES: Data on individual country production are from Jaskula, 2024. List of major lithium miners is from Sean Ashcroft, "Top 10: Lithium Mining Companies," *Mining Digital*, April 3, 2024.

[a] U.S. output was estimated at 2 percent of global production in 2022 (Patrick Whittle, "U.S. Seeks New Lithium Sources as Demand for Clean Energy Grows," *PBS News*, March 28, 2022.)

[b] Jack Conness, "2024 Could Be the Year for American Lithium," *Forbes*, April 14, 2024.

[c] Total outputs summed from individual country totals plus estimates of U.S. output.

Although many producers refine their own lithium ore or brine, the refined products—lithium carbonate or lithium hydroxide—do not attain the purity levels needed for batteries. Refined lithium carbonate, for example, is frequently sold at 99.5 percent purity, but battery manufacturers prefer 99.9 percent purity or higher. China has a dominant position in processing lithium to battery-grade levels and producing global lithium-ion batteries. It reportedly controls 58 percent of global lithium chemical processing capacity that brings lower-purity lithium carbonate or lithium hydroxide up to manufacturers' desired levels.[40] Much of this lithium goes into lithium-ion batteries manufactured in China; China produces over 60 percent of the lithium-ion batteries in the world. China's Contemporary Amperex Technology Co., Limited (CATL) is the largest battery producer in the world, with 34 percent of the global market; it is followed by another Chinese company, BYD Co., with 15.8 percent of the market. Four other Chinese manufacturers rank among the indus-

[40] Alex Scott, "Challenging China's Dominance in the Lithium Market," *Chemical and Engineering News*, Vol. 100, No. 38, October 29, 2022.

try leaders, although Korean and Japanese manufacturers still play important roles in the global market, especially outside China. Korea's LG has 13.6 percent of the global market, followed by Japan's Panasonic and another Korean firm, Samsung.[41]

Chinese Policies

The Chinese government has prioritized the development of EVs and batteries. In 2001, the Ministry of Science and Technology issued the "Major Science and Technology Special Project for Electric Vehicles" under the 863 Plan.[42] The project heavily funded R&D on batteries.

In 2012, the State Council issued the "Energy-Saving and New Energy Vehicle Industry Development Plan (2012–2020)" as part of the Strategic Emerging Industries program.[43] The plan set goals to increase power density and reduce costs by 2020. It called for the creation of regional clusters for the EV battery industry, as well as the development of two or three highly competitive battery companies and two or three world-class manufacturers of anodes, cathodes, electrolytes, and separators.[44] Both the 863 Plan and the Strategic Emerging Industries program channeled very large sums of money into R&D on batteries.

In 2015, under its "Made in China 2025" strategy, the Chinese government again listed lithium-ion batteries as one of ten key technologies. By 2020, Chinese battery and battery component manufacturers were to supply 80 percent of the domestic market; by 2025, the industry was to develop large-scale export capabilities. As part of this effort, the National Power Battery Innovation Center was set up in Beijing in 2016, one of five such major research centers.[45]

Although R&D funding was important, the most important driver of the development of the Chinese EV battery industry was subsidies for the purchase of EVs. In addition to the subsidies, owners of EVs have been permitted to drive in downtown areas of major cities without restrictions. By contrast, diesel- and gasoline-powered cars are restricted to specific days. Only EVs equipped with batteries that meet specific technological targets concerning energy output were eligible to be listed in the "Recommended Model Catalog for the Promotion and Application of New Energy Vehicles," a precondition for receiving subsidies.[46] The

[41] José Pontes, "Top 10 Battery Producers in the World—2023 (Provisional Data)," Clean Technica, January 19, 2024.

[42] Huiwen Gong and Teis Hansen, "The Rise of China's New Energy Vehicle Lithium-Ion Battery Industry: The Coevolution of Battery Technological Innovation Systems and Policies," *Environmental Innovation and Societal Transitions*, Vol. 46, March 2023.

[43] Gong and Hansen, 2023.

[44] Gu Ruizhen [顾瑞珍] and Wu Jing [吴晶], "The 11th Five-Year Plan '863 Plan' Energy-Saving and New Energy Vehicle Project Passed the Acceptance" ["十一五'863计'划节能与新能源汽车项目通过验收"], Xinhua, September 14, 2012.

[45] Gong and Hansen, 2023.

[46] Ministry of Industry and Information Technology, "Automotive Power Battery Industry Specification Conditions" ["汽车动力蓄电池行业规范条件"], Announcement No. 22 of 2015, March 24, 2015.

policy discriminated in favor of domestic EV manufacturers; no foreign manufacturer was ever listed. To ensure eligibility for the subsidies, Chinese automakers shifted battery orders from Korean and Japanese suppliers to domestic manufacturers. As a method of encouraging Chinese manufacturers to improve their technologies and reduce costs in an effort to develop export markets, the subsidy was reduced each year while technical requirements (such as energy density and range) were increased.[47] The catalog was terminated in 2019 when the government decided that the subsidies were no longer needed to develop the industry (although preferences for EVs to access downtown areas of major cities continued). By that time, the Chinese government had spent an estimated $60 billion on subsidies for EVs and the battery industry,[48] and Chinese battery manufacturers had firmly established themselves as leaders in the domestic and global markets.[49]

U.S. Policy Response

The United States has engaged in several initiatives to reduce global reliance on Chinese battery companies by supporting the creation of alternative sources of supply of highly purified lithium. In June 2022, the United States, Australia, Canada, Finland, France, Germany, Japan, the Republic of Korea, Sweden, the United Kingdom, and the European Commission announced the creation of the Minerals Security Partnership.[50] The Partnership was created to bolster supply chains for critical minerals, including lithium. The partnership directly addresses four major critical minerals challenges:

- diversifying and stabilizing global supply chains
- investing in those supply chains
- promoting high environmental, social, and governance standards in the mining, processing, and recycling sectors
- increasing recycling of critical minerals.

The U.S. government has provided support for investments in domestic lithium mining and refining operations from both the IRA and BIL. Investments in other parts of the supply chain for EV batteries are also eligible for funding. In addition to grants, the IRA provides tax incentives to encourage sourcing battery materials from and manufacturing EVs in North American and U.S.-partner countries. The U.S. government has administered grants through the Department of Defense (DoD) under DPA Title III authorities to support domestic lith-

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47 Ministry of Industry and Information Technology, 2015.

48 Amit Katwala, "The World Can't Wean Itself Off Chinese Lithium," *Wired*, June 30, 2022.

49 Gong and Hansen, 2023.

50 U.S. Department of State, "Minerals Security Partnership," webpage, undated.

ium mining and production. It has used both grants and loans administered through the Department of Energy (DOE) to do the same.[51]

These efforts are coordinated through the administration's American Battery Material Initiative, which aligns and leverages federal resources for expanding the end-to-end battery supply chain; works with stakeholders, allies, and partners to develop more-sustainable, secure, and resilient supply chains; helps guide research, grants, and loans supporting environmentally responsible critical minerals extraction, processing, and recycling; and supports faster and fairer permitting for projects that build the domestic supply chain. The initiative is led by a White House steering committee and coordinated by DOE with support from the Department of the Interior.[52]

To discourage purchases of Chinese batteries, the United States increased tariffs on lithium-ion batteries imported from China by 7.5 percentage points, from 3.4 percent to 10.9 percent in 2018.[53] In December 2023, the U.S. government issued regulations stating that vehicles produced by a company with strong ties to China or that contain Chinese-made battery components would be ineligible for EV purchase subsidies. The regulation also excludes subsidies for EVs with batteries that incorporate minerals imported from China, such as lithium, although that part of the regulation will not be enforced until 2025.[54]

Some imports of lithium-ion batteries have been halted by U.S. customs under the provisions of the Uyghur Forced Labor Prevention Act because the batteries were suspected of having been manufactured using forced Uyghur labor in battery plants in the Xinjiang Uyghur Autonomous Region.[55]

Effectiveness of U.S. Policy Measures

U.S. policies have yet to diminish China's dominance of the lithium-ion battery market, although many U.S. government–subsidized investments by private industry in lithium mining, refining, and the production of batteries are underway. As of 2023, China accounted for 65.1 percent of total U.S. imports of EVs batteries, down marginally from 66.2 percent in 2022.[56] China held a 70.4 percent share in total U.S. imports of lithium-ion batteries of all types in 2023, an all-time high.

[51] U.S. Department of Defense, "Defense Production Act Title III Presidential Determination for Critical Materials in Large-Capacity Batteries," press release, April 5, 2022.

[52] White House, "Biden-Harris Administration Driving U.S. Battery Manufacturing and Good-Paying Jobs," fact sheet, October 19, 2022d.

[53] Antonio J. Rivera, James Kim, David R. Hamill, and Birgit Matthiesen, "Section 301 Four-Year Review: A Black Box for the EV Supply Chain," *National Law Review*, January 25, 2023.

[54] Michelle Toh, "New US Rules on Chinese Batteries Could Push Up Price of Electric Cars," CNN, December 4, 2023.

[55] Nichola Groom, "EV Battery Imports Face Scrutiny Under US Law on Chinese Forced Labor," Reuters, August 19, 2023.

[56] U.S. Census Bureau, undated-a.

The denial of purchase subsidies for EVs that use batteries that incorporate minerals imported from China has had several effects. After congressional hearings and a reappraisal of the future trajectory for growth in EVs, in September 2023, Ford halted construction of a $3.5 billion battery factory in Michigan that was to use technology from CATL.[57] Tesla, General Motors, and Ford, which use Chinese batteries in some models, saw their IRA vehicle purchase subsidies for those models halve in 2024, giving them an impetus to diversify to non-Chinese suppliers.[58]

Uptake of U.S. subsidies for lithium mining, refining, and battery and battery component manufacturing has been strong. In 2022, DOE provided $2.8 billion in funding for 20 manufacturing and processing companies for lithium-ion batteries and other battery material from BIL.[59] Among the projects was a $2.26 billion loan to Lithium America's Thacker Pass project in Nevada for an on-site refining facility.[60] Other recipients include Albemarle Corporation, which received $150 million to build a new lithium processing plant in Kings Mountain, North Carolina, and Piedmont Lithium, which plans to build new lithium refining plants in North Carolina and Tennessee.[61] A large number of other, smaller companies are developing new technologies to mine and refine lithium. Of particular note are companies that are developing technologies for direct lithium extraction, which could be cheaper than existing refining technologies while being more environmentally friendly.[62] On November 15, 2023, DOE announced that a second tranche of $3.5 billion in funding had been released. Funding for battery projects from BIL is to total $6 billion.[63] In light of the number of projects and the volume of investment going into new mines and refineries, U.S. dependence on China for lithium and lithium-ion batteries should fall significantly over the next several years.

Case Study 3: Graphite

Graphite is an allotrope of carbon that has several manufacturing applications, including in batteries for EVs and in foundries, lubricants, and metallurgy. Graphite is the largest component in lithium-ion batteries; the anode material consists of 95 to 99 percent graph-

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57 Kirsten Korosec, "Ford Halts Work on $3.5B EV Battery Factory with China's CATL," *TechCrunch*, September 25, 2023.

58 David Nadelle, "7 EVs and Hybrid Cars That No Longer Qualify for a Tax Credit in 2024," GOBanking Rates, April 22, 2024.

59 White House, 2022d.

60 Conness, 2024.

61 Pippa Stevens, "Miner Piedmont Unveils Plans to Build Lithium Refining Plant in Push for Domestic EV Supply Chains," CNBC, September 1, 2022.

62 Rakesh Krishnamoorthy Iyer and Jarod C. Kelly, *Lithium Production in North America: A Review*, ANL/ESIA-23/8, Argonne National Laboratory, October 1, 2023.

63 U.S. Department of Energy, "Biden-Harris Administration Announces $3.5 Billion to Strengthen Domestic Battery Manufacturing," November 15, 2023b.

ite. Each EV requires about 110 to 220 pounds of graphite in its battery pack's anodes.[64] This case study focuses specifically on graphite applications in the lithium-ion battery supply chain.

Commercially sold graphite comes in two forms: natural and synthetic. Natural graphite is mined; synthetic graphite is produced using high temperature treatment of carbon compounds left over from refining crude oil. The United States manufactures synthetic graphite but does not mine natural graphite. Synthetic graphite is more desirable in manufacturing applications because of its higher purity levels, but it is more expensive to produce than natural graphite, and the manufacturing process typically emits more greenhouse gases.[65] U.S. lithium-ion batteries are usually made with synthetic graphite supplemented with natural graphite, but about 80 percent of the graphite used in lithium-ion batteries globally is natural.[66] In recent years, rising synthetic graphite prices have led to increased market share for natural graphite in battery anode production, although synthetic graphite continues to dominate the market in the United States.[67]

Trends in Global Market Share

Global natural graphite production ran 1.68 million and 1.6 million tons in 2022 and 2023, respectively.[68] Synthetic graphite production runs about 1.8 million tons.[69] China dominates the global graphite market, producing approximately 70 percent of all synthetic and natural graphite. It also produces over 90 percent of the anodes used in lithium-ion batteries globally.[70] China has ramped up production of synthetic graphite even as it has reduced natural graphite mining because of environmental issues.[71]

Demand for graphite used in battery anodes has outstripped supply. In 2022, anode demand grew by 46 percent, while the supply of flake graphite grew by only 14 percent.

[64] Paul Lienert and Nick Carey, "Focus: Synthetic Graphite for EV Batteries: Can the West Crack China's Code?" Reuters, September 13, 2023.

[65] Zhang Jinrui, Chao Liang, and Jennifer B. Dunn, "Graphite Flows in the U.S.: Insights into a Key Ingredient of Energy Transition," *Environmental Science and Technology,* Vol. 57, No. 8, February 15, 2023, pp. 3402–3403.

[66] Zhang, Liang, and Dunn, 2023, p. 3403.

[67] Jon Stibbs and Sybil Pan, "Synthetic Versus Natural Graphite Debate Rages On: 2023 Preview," Fastmarkets, January 17, 2023.

[68] U.S. Geological Survey, 2024, p. 85.

[69] ReportLinker, "The Global Synthetic Graphite Market Was at 1,814.61 Kilotons in 2021 and the Market Is Projected to Register a CAGR of over 4.21% During the Forecast Period (2022-2027)," GlobeNewswire, September 27, 2022.

[70] Hannah Northey and Kyle Duggan, "US and Canada Confront the Next Big EV Minerals Challenge," *E&E News,* November 30, 2023.

[71] Siyi Liu and Dominique Patton, "China, World's Top Graphite Producer, Tightens Exports of Key Battery Material," Reuters, October 20, 2023.

Graphite prices have increased in recent years in response to rising demand.[72] Flake graphite prices rose by 25 percent in 2022, for example.[73] In 2023, global sales of EVs and hybrid vehicles rose by 30 percent.[74] As global demand for EVs and hybrid vehicles continues to rise, analysts project that demand for graphite will continue to grow as well.

The United States, the EU, and South Korea are the largest importers of graphite, synthetic and natural.[75] The United States imports the plurality (33 percent) of its graphite from China but also imports significant shares from Mexico (18 percent), Canada (17 percent), and Madagascar (10 percent).[76] In 2023, there were ten lithium-ion battery manufacturing facilities in the United States, with an additional 28 under development.

Chinese Policies

In October 2023, China placed restrictions on exports of graphite. The move was seen as a response to the Biden administration's efforts to block semiconductor exports to China and measures under both the Trump and Biden administrations to levy tariffs on solar energy equipment manufactured in China.[77] China announced the new graphite export restrictions three days after the United States released new controls on exports of high-end semiconductors. China's restrictions are country agnostic and classify graphite as a dual-use item that requires a special export license. Chinese officials can decide whether to block specific requests for exports.[78] Analysts saw the special license requirement as "a geopolitical signaling device" intended to demonstrate that China is willing to impose further economic restrictions, not as a tool for retaliation per se.[79] Because China did not impose a blanket ban on graphite exports, analysts say it will be difficult to estimate the policy's potential effects on the global market.[80]

72 Ashutosh Pandey, "China's Graphite Dominance Threatens Electric Car Market," *Deutsche Welle,* March 14, 2022.

73 George Miller, "What to Expect for Graphite in 2023?" *Benchmark Source*, Benchmark Mineral Intelligence, January 11, 2023.

74 Nick Carey, "Global Electric Car Sales Rose 31% in 2023—Rho Motion," Reuters, January 10, 2024.

75 Alexander C. Kaufman, "China Restricts Exports of a Key Mineral, Stoking U.S. Fears About Battery Supply Chains," *HuffPost*, October 20, 2023.

76 U.S. Geological Survey, *Mineral Commodity Summaries 2023*, U.S. Department of the Interior, 2023, p. 82.

77 Kaufman, 2023.

78 Lily Kuo, "The Next Front in the Tech War with China: Graphite (and Clean Energy)," *Washington Post,* November 29, 2023.

79 Emily Benson and Thibault Denamiel, "China's New Graphite Restrictions," Center for Strategic and International Studies, October 23, 2023.

80 Liu and Patton, 2023.

U.S. Policy Response

U.S. policy measures to increase domestic production of graphite while reducing U.S. reliance on imports of Chinese graphite are part of the Biden administration's broader efforts to combat climate change and secure renewable energy supply chains. These policies include an emphasis on transitioning from vehicles powered by fossil fuels to EVs and shoring up the relevant EV supply chains in the United States and alongside international allies and partners. The Biden administration has set a goal that 50 percent of all vehicles sold in the United States will be EVs by 2030.[81]

In 2021, a consortium of U.S. federal agencies set the goal of building a secure lithium battery material and technology supply chain in the United States or with U.S. partners by 2030. Elements of achieving this goal include "secur[ing] access to raw and refined materials" and "support[ing] the growth of a U.S. materials-processing base able to meet domestic battery manufacturing demand."[82] In June 2022, the United States, 13 other countries, and the EU announced the Minerals Security Partnership "to facilitate targeted financial and diplomatic support for strategic projects along the value chain."[83]

The United States has pursued two main avenues to achieve these objectives: grants under DPA Title III authorities and grants and loans administered by DOE to support domestic mining and production. In March 2022, the Biden administration issued a determination "permitting the use of DPA Title III authorities to strengthen the U.S. industrial base for large-capacity batteries" and allowing DoD "to increase domestic mining and processing of critical materials for the large-capacity battery supply chain."[84] The determination directed the Secretary of Defense to "create, maintain, protect, expand, or restore sustainable and responsible domestic production capabilities of . . . strategic and critical materials."[85] As part of this effort, DoD was directed to conduct feasibility studies on the locations of graphite reserves, which could increase investor interest and reduce the risks involved for investors in future mining and refining projects.[86]

Additionally, the IRA includes provisions for funding grants to be allocated by DoD under Title III of the DPA. In July 2023, DoD announced a grant of $37.5 million for Graphite One

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[81] White House, "President Biden Announces Steps to Drive American Leadership Forward on Clean Cars and Trucks," fact sheet, August 5, 2021b.

[82] Federal Consortium for Advanced Batteries, *National Blueprint for Lithium Batteries: 2021-2030*, executive summary, Vehicle Technologies Office, Office of Energy Efficiency and Renewable Energy, U.S. Department of Energy, June 2021.

[83] U.S. Department of State, undated.

[84] U.S. Department of Defense, 2022.

[85] Presidential Determination No. 2022-16, "Memorandum on Presidential Determination Pursuant to Section 303 of the Defense Production Act of 1950, as Amended, on Insulation," Executive Office of the President, June 6, 2022.

[86] Joshua Busby, Emily Holland, Morgan Bazilian, and Paul Orszag, "The Defense Production Act's Role in the Clean Energy Transition," *Lawfare,* July 17, 2023.

to perform an accelerated feasibility study for its Graphic Creek natural graphite project in Washington state. The project plan includes a manufacturing plant for advanced graphite and battery anodes and a battery material recycling facility.[87] The U.S. Geological Survey identified the Graphite Creek deposit as "among the largest in the world."[88] Graphite One aims to finish its commercial facility in late 2025 and to produce 25,000 tons of anode material by mid-2026.[89] In November 2023, also under DPA Title III authorities, DoD announced a $3.2 million grant to South Star Battery Metals Corporation, a Canadian company, to support a feasibility study for the production of purified natural graphite at a project located in Alabama.[90]

BIL also includes provisions for investments administered by DOE in the domestic battery industry. The law includes a goal to develop enough battery-grade graphite for 1.2 million EVs on an annual basis.[91] Funded by BIL, DOE awarded $2.8 billion in grants related to the domestic battery industry. Additionally, Syrah Technologies received a $102 million loan from DOE, announced in July 2022, to expand the capacity of its synthetic graphite anode material facility in Vidalia, Louisiana. The Syrah Vidalia facility aims to be "the only vertically integrated, large-scale" manufacturer of active anode material outside China and would be "the first of its kind in the United States."[92] In the first quarter of 2024, the Vidalia facility had a production capacity of 11.25 metric tons per annum of active anode material; a feasibility study confirmed that an expansion of the facility's production capacity to 45,000 metric tons per annum is technically and financially feasible.[93] NOVONIX, a battery technology company with operations in the United States and Canada, has received a $150 million grant to expand its production of synthetic graphite, with the goal of expanding production to 40,000 metric tons per annum by 2025.[94] Anovion Battery Materials received $117 million to expand its manufacturing capacity of synthetic graphite materials to 35,000 metric tons per annum at a

[87] U.S. Department of Defense, "DOD Enters Agreement to Expand Capabilities for Domestic Graphite Mining and Processing for Large-Capacity Batteries," press release, July 17, 2023a.

[88] Graphite One, "New U.S. Government Report Identifies Graphite One's Graphite Creek Deposit 'Is Among the Largest in the World," March 9, 2023.

[89] Graphite One, "Graphite One 2023 Year in Review," January 2, 2024.

[90] U.S. Department of Defense, "DOD Enters Agreement to Expand Domestic Graphite Supply Chain," press release, November 29, 2023b.

[91] Jon Stibbs, "US Department of Energy Issues $2.8bln in Grants for Domestic Battery Industry," Fastmarkets, October 21, 2022.

[92] Loan Programs Office, "Syrah Vidalia," webpage, U.S. Department of Energy, undated.

[93] Syrah Resources, "Vidalia Active Anode Material Facility," webpage, undated.

[94] NOVONIX, "NOVONIX Selected for US$150 Million Grant from U.S. Department of Energy," October 20, 2022.

plant in New York state.[95] Four other companies have received grants focused on replacing the graphite component of battery anodes with silicon.[96]

Effectiveness of U.S. Policy Measures

Government programs to provide subsidies and other support for domestic investments in graphite mining and production appear to have been successful, furthering the diversification of U.S. supplies. Several companies are moving forward with investments in natural graphite mining and synthetic graphite production capacity. In 2018, the United States consumed a total of about 354,000 metric tons of synthetic and natural graphite combined.[97] U.S.-based companies will therefore need to scale up their production considerably to meet demand. However, the United States may be able to work with allies and partners to increase supplies of graphite from countries other than China. Brazil, Madagascar, Mozambique, and Turkey all have considerable natural graphite reserves (Table 3.3). Advanced graphite mining projects are also under development in Sweden, Canada, and Australia.[98]

Case Study 4: Rare Earth Minerals

Rare earth minerals are a set of 17 metallic elements that are critical elements in components for smart phones, EVs, wind turbines, defense technologies, and other products.[99] Despite their name, rare earths are not rare; rather, they "are relatively abundant in the Earth's crust, but minable concentrations are less common than for most other mineral commodities," according to the U.S. Geological Survey.[100] Once rare earth minerals are mined, they are refined to convert the ore into the metals and oxides that are used in downstream products.[101]

Trends in Global Market Share

In 2023, China produced 68 percent of global rare earths and had over 85 percent of rare earth mining capacity (Table 3.4). It manufactured more than 90 percent of the global production of magnets made from rare earths, which are used in wind turbines, electric motors,

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95 Anovian Technologies, "Anovion Technologies Selected to Receive $117 Million Grant Under the Bipartisan Infrastructure Law for Battery Materials Processing and Manufacturing," press release, October 19, 2022.

96 Stibbs, 2022.

97 Zhang, Liang, and Dunn, 2023.

98 U.S. Geological Survey, 2023, p. 84.

99 Rare earths include the 15 lanthanides found on the periodic table, plus scandium and yttrium (American Geosciences Institute, "What Are Rare Earth Elements, and Why Are They Important?" webpage, undated).

100 U.S. Geological Survey, *Mineral Commodity Summaries 2024*, U.S. Department of the Interior, January 31, 2024.

101 Matt Blois, "Firms Aim to Boost Rare Earth Processing in the US," *Chemical and Engineering News,* August 10, 2023.

TABLE 3.3
2023 Global Natural Graphite Mine Production and Reserves

Country	Mine Production (metric tons)	Reserves (metric tons)
Austria	500	N/A[a]
Brazil	73,000	74,000,000
Canada	3,500	5,700,000
China	1,230,000	78,000,000
Germany	150	N/A[a]
India	11,500	8,600,000
Madagascar	100,000	24,000,000
Mexico	2,000	3,100,000
Mozambique	96,000	25,000,000
North Korea	8,100	2,000,000
Norway	7,200	600,000
Russia	16,000	14,000,000
South Korea	27,000	1,800,000
Sri Lanka	2,200	1,500,000
Tanzania	6,000	18,000,000
Turkey	2,000	6,900,000
Ukraine	2,000	N/A[a]
Vietnam	500	N/A[a]
World total	1,600,000	280,000,000[a]

SOURCE: Data are from U.S. Geological Survey, *Mineral Commodity Summaries 2024*, U.S. Department of the Interior, 2024, p. 85.

NOTE: N/A = Not available; the U.S. Geological Survey did not report reserves.

[a] Countries without reserves listed are included in the world total.

and other applications.[102] China has accounted for most of the world's rare earth production since the early 2000s. There are several major rare earth mining sites outside China, however, including the Lynas Rare Earths facilities in Australia and MP Materials in the United States and Canada.[103] MP Materials operates the only rare earth mine in the United States in Mountain Pass, California. Mining at Mountain Pass produces approximately 12 percent of

102 Mikayla Easley, "Special Report: U.S. Begins Forging Rare Earth Supply Chain," *National Defense,* February 10, 2023.

103 Maddie Stone, "Offshore Wind Turbines Need Rare Earth Metals. Will There Be Enough to Go Around?" *Grist,* October 6, 2023b.

TABLE 3.4

Global Rare Earth Mine Production and Reserves in 2023

Country	Output (in metric tons rare-earth-oxide equivalent)	Output as a Percentage of Global Total	Reserves (in metric tons rare-earth-oxide equivalent)
China	240,000	67.93	44,000,000
United States	43,000	12.17	1,800,000
Burma	38,000	10.76	N/A
Australia	18,000	5.09	5,700,000
Thailand	7,100	2.01	4,500
India	2,900	0.82	6,900,000
Russia	2,600	0.74	10,000,000
Madagascar	960	0.27	N/A
Vietnam	600	0.17	22,000,000
Brazil	80	0.02	21,000,000
Malaysia	80	0.02	N/A
Greenland	N/A	N/A	1,500,000
Tanzania	N/A	N/A	890,000
Canada	N/A	N/A	830,000
South Africa	N/A	N/A	790,000
World Total	353,320	100.00	115,000,000[a]

SOURCE: Data are from U.S. Geological Survey, 2024, p. 145.

[a] Calculated from listed countries with reported reserves. Only the reserve world total is rounded.

the global output of rare earth concentrate (Table 3.4).[104] Between 2019 and 2022, the United States received 72 percent of its rare earth imports from China, 11 percent from Malaysia, 6 percent from Japan, 5 percent from Estonia, and 5 percent from other countries.[105]

Many countries have potentially commercially exploitable reserves of rare earths (Table 3.4). Several countries in Africa have significant rare earth mining potential, but "Africa's full potential in rare earths is largely untapped given low levels of exploration" on the continent.[106]

[104] Yusuf Khan, "The U.S. Wants a Rare-Earths Supply Chain. Here's Why It Won't Come Easily," *Wall Street Journal,* April 25, 2023.

[105] U.S. Geological Survey, 2024.

[106] Gracelin Baskaran, "Could Africa Replace China as the World's Source of Rare Earth Elements?" Brookings Institution, December 29, 2022.

Chinese Policies

China dominates the global rare earth supply chain because, since the early 2020s, the Chinese government has invested heavily in rare earth mining and processing despite the environmentally destructive effects of this industry; the chemicals used to extract rare earth minerals from ore create toxic waste that contaminates soil, waterways, and air.[107] China moved to restrict rare earth exports for the first time in 2010 to punish Japan over a dispute regarding the Senkaku Islands.[108] China's share of global rare earth production subsequently fell from over 95 percent in 2010 to 70 percent in 2019.[109] It has continued to use its dominant position in rare earth refining technologies as an economic and foreign policy instrument. In December 2023, China banned exports of technologies for processing rare earths and for manufacturing some rare earth magnets in response to U.S. export restrictions on advanced semiconductors.[110]

U.S. Policy Response

Recent U.S. administrations have identified rare earths as key to U.S. defense and economic interests. In 2018, rare earths were included on a list of more than 50 critical materials released by the U.S. government; the list was updated in 2022, as directed by the Energy Act of 2020.[111] In June 2021, the Biden administration issued an assessment of U.S. supply chain vulnerabilities, finding that "over-reliance on foreign sources and adversarial nations for critical minerals and materials posed national and economic security threats."[112]

The U.S. government provides tax credits through the IRA for domestic producers of critical minerals, including some rare earth elements. It also provides $500 million under the DPA for economic incentives for the extraction and refining of critical minerals and $40 million in DOE loan guarantees, including for projects that "'increase the domestically produced supply of critical minerals.'"[113] DoD has announced several grants from these funds for rare

107 Jaya Nayar, "Not So 'Green' Technology: The Complicated Legacy of Rare Earth Mining," *Harvard International Review,* August 12, 2021.

108 Keith Bradsher, "Amid Tension, China Blocks Vital Exports to Japan," *New York Times*, September 22, 2010.

109 "Rare Earths Give China Leverage in the Trade War, at a Cost," *The Economist*, June 15, 2019.

110 Milton Ezrati, "How Much Control Does China Have over Rare Earth Elements?" *Forbes*, December 11, 2023.

111 The Energy Act of 2020 defines *critical materials* as "any non-fuel mineral, element, substance, or material that the Secretary of Energy determines: (i) has a high risk of supply chain disruption; and (ii) serves an essential function in one or more energy technologies, including technologies that produce, transmit, store, and conserve energy" (U.S. Department of Energy, "What Are Critical Materials and Critical Minerals?" Critical Minerals and Materials Program, webpage, undated).

112 White House, "Securing a Made in America Supply Chain for Critical Minerals," fact sheet, February 22, 2022a.

113 Oscar Serpell, "Impacts of the Inflation Reduction Act on Rare Earth Elements," Kleinman Center for Energy Policy, September 24, 2022.

earths, including a $9.6 million grant to MP Materials. In November 2020, DoD awarded two grants to Lynas Rare Earths worth $150 million. In February 2022, DoD announced an additional $35 million grant to MP Materials from its Industrial Base Analysis and Sustainment Program. In April 2023, DOE announced $16 million in grants funded from BIL for R&D on rare earth refineries. And in August 2023, DOE announced the availability of up to $30 million in funding aimed at reducing U.S. reliance on imported rare earths.[114]

The Biden administration has also increased funding for research, including $74 million annually from 2022 to 2026 from BIL for the U.S. Geological Survey's Earth Mapping Resources Initiative.[115] Less than 40 percent of the United States is currently mapped at a great enough level of detail to support the discovery of rare earth deposits. Once potential mining sites have been mapped, private companies are in a position to conduct more highly detailed exploratory work. This "can take an additional several years, after which it might take up to a decade to permit and build a mine."[116] A research project from the Defense Advanced Research Projects Agency is also investigating the potential use of microbial and biomolecular engineering to process rare earths.[117]

Effectiveness of U.S. Policy Measures

It is difficult to measure the success of these initial investments in domestic rare earth mining and processing at present, given that many of these investments will require several years before they result in new production. MP Materials, the only company that currently mines rare earths in the United States, has said that the company's goal is to "build a full magnetic supply chain," and the company "want[s] to be able to make all the necessary materials and recycle the necessary materials to have that magnetic supply chain."[118] As of 2023, while MP Materials mined rare earth materials in the United States, the ore was still shipped to China for refining. However, the company is close to completing a refinery, for which it received a $35 million contract from DoD in February 2022. The company is also building the first U.S. rare earth magnet factory in Fort Worth, Texas.[119] Other U.S. companies are planning on opening mines and potentially refineries as well, which will significantly expand production outside China.[120]

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114 U.S. Department of Energy, "Biden-Harris Administration Announces $30 Million to Build Up Domestic Supply Chain for Critical Minerals," August 21, 2023a.

115 The program also supports mapping of other critical mineral deposits, such as lithium.

116 Maddie Stone, "A Government Program Hopes to Find Critical Minerals Right Beneath Our Feet," *Grist,* March 17, 2023a.

117 Easley, 2023.

118 Easley, 2023.

119 Easley, 2023.

120 "American Rare Earths Boosts Tonnage at Halleck Creek Project in Wyoming," Mining.com, February 7, 2024

Case Study 5: Advanced Semiconductors

Advanced semiconductors have critical applications in artificial intelligence (AI) and defense platforms and systems. U.S. officials have therefore deemed maintaining an edge in advanced semiconductor development and manufacturing a national security concern. The United States has taken a variety of measures, including export controls and investments in the domestic advanced semiconductor industry, to maintain its edge.

Trends in Global Market Share

Advanced semiconductor supply chains are globalized, but several steps in the manufacturing process—from design to the manufacture of chips at a foundry—are limited to a small number of producers in just a few countries because of the high production costs and specialized equipment and knowledge needed to undertake them. As of 2022, the top nine global semiconductor manufacturing equipment producers by revenue were

1. Advanced Semiconductor Materials Lithography (ASML; Netherlands)
2. Applied Materials (United States)
3. Lam Research (United States)
4. Tokyo Electron (Japan)
5. KLA (United States)
6. Advantest (Japan)
7. Teradyne (United States)
8. SCREEN SPE (Japan)
9. Hitachi Hi-Tech (Japan).[121]

Three countries (the United States, the Netherlands, and Japan) control more than 90 percent of the production of manufacturing equipment that is used to build advanced semiconductors. A Dutch company, ASML, is the only producer of extreme ultraviolet lithography machines, which are required to manufacture the most-advanced chips.[122] As a consequence, the advanced semiconductor manufacturing equipment industry is characterized by chokepoints in the global supply chain that provide an effective means of implementing export controls and other measures aimed at denying China access to the most-advanced chips.

China is a major consumer of semiconductors, including (until the recent export ban) advanced semiconductors. China plus Hong Kong accounted for 51 percent of global imports of semiconductors in 2022 (Table 3.5). They are also important exporters, accounting for

[121] Andre Barbe and Will Hunt, "Preserving the Chokepoints: Reducing the Risk of Offshoring Among U.S. Semiconductor Manufacturing Equipment Firms," policy brief, Center for Security and Emerging Technology, May 2022, pp. 3–4.

[122] Katie Tarasov, "ASML Is the Only Company Making the $200 Million Machines Needed to Print Every Advanced Microchip. Here's an Inside Look," CNBC, March 23, 2022.

TABLE 3.5
Imports and Exports of Semiconductors, by Country in 2022

Country	Imports ($ billions)	Imports (%)	Country	Exports[a] ($ billions)	Exports (%)
China and Hong Kong	638.7	50.9	China and Hong Kong	368.3	34.2
Singapore	108.0	8.6	Taiwan	183.7	17.1
Taiwan	87.5	7.0	Singapore	122.0	11.3
South Korea	62.4	5.0	South Korea	112.8	10.5
Vietnam	54.3	4.3	Malaysia	78.6	7.3
Malaysia	53.5	4.3	United States	51.6	4.8
United States	43.7	3.5	Japan	33.6	3.1
Japan	31.7	2.5	Philippines	28.9	2.7
Mexico	26.3	2.1	Germany	19.9	1.8
Germany	23.8	1.9	Vietnam	13.3	1.2
World total[b]	1,255.9			1,076.6	

SOURCE: Data are from UN Comtrade Database, "Trade Data," United Nations, undated.

NOTE: This table includes only the top importers and exporters of semiconductors.

[a] Exports do not equal imports because imports include costs, insurance, and freight.

[b] The world totals include import and export values from other countries not listed here.

34 percent of the global total. However, China plus Hong Kong import 3.4 times more than they export, primarily because semiconductors are incorporated into so many important Chinese exports.

Chinese Policies

China has long held ambitions to develop an indigenous, vertically integrated semiconductor industry. In 2014, the government published a plan with the goal of "establishing a world-leading semiconductor industry in all areas of the integrated circuit supply chain by 2030."[123] Since then, China has been investing heavily in its domestic semiconductor manufacturing capacity. The "Made in China 2025" industrial strategy aims to produce 80 percent of China's domestic consumption of semiconductors by 2030. To meet that goal, the Chinese government has provided an estimated $150 billion worth of subsidies to its chip industry over the past decade.[124]

Although China currently lacks the ability to manufacture cutting edge chips, it is using restrictions on exports of key materials to potentially control the manufacture of chips

[123] Karen M. Sutter, *China's New Semiconductor Policies: Issues for Congress*, Congressional Research Service, R46767, April 20, 2021, pp. 3–4.

[124] "China Is Quietly Reducing Its Reliance on Foreign Chip Technology," *The Economist,* February 13, 2024.

elsewhere in the world. In July 2023, it announced restrictions on exports of gallium and germanium, metals that are used to manufacture some semiconductor wafers. The new regulations require exporters to apply for a license. However, China has not banned specific end-uses or importers.[125]

U.S. Policy Response

U.S. officials have said they are taking a "small yard, high fence" approach—that is, aggressively protecting critical technologies while allowing most economic relations with China to proceed as usual. For semiconductors, this has meant preventing China from accessing "advanced semiconductor manufacturing tools, the most advanced chips, and supercomputing capabilities."[126] The United States has made use of several measures in recent years to limit China's access to advanced semiconductors and shore up friendly supply chains. These measures include imposing export controls in collaboration with international partners, adding the Chinese company Huawei to the Entity List, levying tariffs, restricting investment by U.S. companies in semiconductor manufacturing companies in China, and investing in the United States' own advanced semiconductor industry.

On October 7, 2022, the Bureau of Industry and Security (BIS), part of the U.S. Department of Commerce, announced export controls on advanced semiconductors destined for China. The rules included restrictions on high-performance chips, products that contain those chips, and the tools used to manufacture them.[127] A year later, in October 2023, U.S. officials announced an update to the rules to close loopholes that China had been using to circumvent the rules.[128]

Effective May 16, 2019, BIS added the Chinese company Huawei and several non-U.S. Huawei affiliates to the Entity List, thereby imposing a licensing requirement on U.S. companies to export to Huawei or its affiliates all items that are subject to the Export Administration Regulations. Effective May 2020, BIS amended the Foreign Direct Product Rule "to further limit technology releases to designated Huawei entities, primarily targeted on the semiconductor industry."[129] In December 2020, BIS added China's Semiconductor Manufacturing

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[125] Clement Tan, "China Slaps Export Curbs on Chipmaking Metals in Tech War Warning to U.S., Europe," CNBC, July 3, 2023.

[126] Jake Sullivan, "Remarks by National Security Advisor Jake Sullivan on the Biden-Harris Administration's National Security Strategy," White House, October 12, 2022.

[127] Bureau of Industry and Security, "Implementation of Additional Export Controls: Certain Advanced Computing and Semiconductor Manufacturing Items; Supercomputer and Semiconductor End Use; Entity List Modification," *Federal Register*, U.S. Department of Commerce, Vol. 87, No. 197, October 13, 2022b.

[128] Bureau of Industry and Security, "Commerce Strengthens Restrictions on Advanced Computing Semiconductors, Semiconductor Manufacturing Equipment, and Supercomputing Items to Countries of Concern," press release, U.S. Department of Commerce, October 17, 2023.

[129] Kay C. Georgi and Sylvia G. Costelloe, "BIS Expands the Huawei Foreign Direct Product Rule to Capture a Wide Swath of COTS Products," ArentFox Schiff, August 19, 2020.

International Corporation (SMIC) to the Entity List.[130] In addition, the Trump administration's 2018 increase in tariffs on Chinese imports under Section 301 authorities included tariffs on some semiconductors manufactured in China.[131]

In August 2023, Biden issued an Executive Order regulating outbound U.S. investments in certain semiconductors, quantum computing, and AI capabilities in China.[132] According to U.S. officials, the regulations were intended to restrict not just the flow of U.S. capital to China but also the transfer of knowledge to Chinese industry.[133]

Signed into law in August 2022, the CHIPS and Science Act authorized $280 billion in funding for domestic research and manufacturing of semiconductors and authorized investment tax credits for the costs of purchasing semiconductor manufacturing equipment.[134] In September 2023, the Biden administration finalized rules that prohibit companies that receive this funding from carrying out business, partnerships, and research in China or from using federal funding to build factories outside the United States.[135]

Effectiveness of U.S. Policy Measures

The export restrictions on semiconductors to China have had measurable effects on Chinese imports of semiconductors. After U.S. and allied export controls were implemented, Chinese imports fell 16 percent in 2023 compared with 2022. Imports from the United States, in particular, fell sharply, down 52 percent in 2023 from their peak in 2021.[136] Chinese imports of semiconductor manufacturing equipment have also fallen.[137] A 2023 report by the U.S. International Trade Commission concludes that the Section 301 tariffs reduced U.S. imports of semiconductors from China by over 70 percent and raised the value of U.S. semiconductor production by about 6 percent.[138] However, according to the U.S. Semi-

[130] Bureau of Industry and Security, "Commerce Adds China's SMIC to the Entity List, Restricting Access to Key Enabling Technology," press release, U.S. Department of Commerce, December 18, 2020.

[131] Andres B. Schwarzenberg, *Section 301 Tariff Exclusions on U.S. Imports from China*, Congressional Research Service, IF11582, May 13, 2024.

[132] Executive Order 14105, "Addressing United States Investments in Certain National Security Technologies and Products in Countries of Concern," Executive Office of the President, August 9, 2023.

[133] Hans Nichols, "Biden's Real Target on Chinese Investment Restrictions," *Axios,* August 9, 2023.

[134] John F. Sargent, Jr., Karen M. Sutter, and Manpreet Singh, *Frequently Asked Questions: CHIPS Act of 2022 Provisions and Implementation*, Congressional Research Service, R47523, April 25, 2023.

[135] Ana Swanson, "U.S. Issues Final Rules to Keep Chip Funds Out of China," *New York Times*, September 22, 2023a.

[136] "China's Imports of ICs Fell in 2022 for First Time Since 2004," Bloomberg, January 13, 2023.

[137] Ailing Tan and James Mayger, "China's Imports of Chip-Making Gear Drop to Lowest Since Mid-2020," Bloomberg, December 21, 2022.

[138] U.S. International Trade Commission, "Certain Effects of Section 232 and 301 Tariffs Reduced Imports and Increased Prices and Production in Many U.S. Industries," News Release 23-024, Inv. No(s). 332-591, March 15, 2023.

conductor Industry Association, the tariffs exacerbated chip shortages caused by supply chain disruptions during the COVID-19 pandemic and added 25 percent to the cost of semiconductors.[139]

Placing Huawei and other Chinese companies on the Entity List has had profound negative consequences for them. Following the addition of Huawei and its non-U.S. affiliates to the Entity List in 2019, Huawei's operations were severely disrupted; annual revenue declined by 30 percent in 2021 and remained flat in 2022. SMIC has benefited from the cutoff in exports of Western chips to China, as Chinese companies have turned to it for substitutes.[140] However, the addition of SMIC to the Entity List has made it more difficult for the company to purchase chip manufacturing equipment.

The export restrictions were designed to freeze Chinese companies at the 14-nanometer threshold. In 2023, a teardown of Huawei's new Mate 60 Pro phone revealed that the product was powered by a 7-nanometer chip made by SMIC. Although officials worried that this news meant that the U.S. export restrictions on chips were ineffective, analysts argued that SMIC had "achieved its 7-nanometer breakthrough months, if not a full year, before the Biden administration imposed export controls on China."[141] Analysts also suggest that SMIC may not be able to manufacture the 7-nanometer chips in large quantities, providing evidence that the export controls are impeding China's development of chip manufacturing capabilities.[142]

Analysts believe the 7-nanometer breakthrough was facilitated because China was able to take advantage of the lag between the initial tranche of U.S. restrictions in October 2022 and the export restrictions announced by the Netherlands and Japan, which took effect in mid-2023 through early 2024. As noted, ASML is the only producer of extreme ultraviolet lithography machines, which are required to manufacture the most-advanced chips.[143] China's imports of lithography machines surged in the lead-up to the Dutch restrictions coming into effect in January 2024.[144]

Over the longer-term, China is reportedly building a domestic chipmaking supply chain to substitute for imported technologies. According to *The Economist*, "China's government is thought to be easing the way by providing subsidies to [chipmakers] that purchase local equipment."[145] Still, Chinese production lags behind the cutting edge of the chipmaking

[139] Jennifer Meng, "SIA Urges Elimination of Harmful Section 301 Tariffs," Semiconductor Industry Association, December 8, 2021.

[140] Arjun Kharpal, "China's Biggest Chipmaker SMIC Posts Record 2022 Revenue but Warns of a Tough Year Ahead," *CNBC,* February 10, 2023.

[141] Megan Hogan, "Export Controls Are Only a Short-Term Solution to China's Chip Progress," *War on the Rocks,* December 22, 2023.

[142] Hogan, 2023.

[143] Tarasov, 2022.

[144] "China Is Quietly Reducing Its Reliance on Foreign Chip Technology," 2024.

[145] "China Is Quietly Reducing Its Reliance on Foreign Chip Technology," 2024.

industry: South Korea's Samsung and the Taiwan Semiconductor Manufacturing Company have been producing 3-nanometer chips since 2022.[146]

The structure of the global advanced semiconductor supply chain and the export restrictions will make it difficult for China to innovate around them, at least for the next five to ten years.[147] Chinese firms are far behind Western semiconductor equipment manufacturers. In photolithography, the sector in which China lags the furthest behind, a 2019 study estimated that it would take China at least a decade to build up a domestic photolithography industry to replace imports that are now blocked.[148] Export restrictions will likely not prevent China from advancing its domestic advanced semiconductor industry over the long run, but they are likely to slow China's progress and raise its costs.

Investigations by CFIUS affected at least five proposed investments in U.S. semiconductor companies between 2017 and 2022, according to one analysis. CFIUS blocked one deal, and two others were abandoned because of the increased scrutiny.[149] Magnachip Corp., a company with operations in South Korea that is listed on the New York Stock Exchange, and Wise Road Capital, a Chinese private equity firm, abandoned plans for a merger with an estimated value of $1.4 billion in December 2021 because of scrutiny by CFIUS. Although Magnachip Corp. argued that CFIUS did not have jurisdiction because most of its assets and employees are outside the United States, CFIUS required the parties to the deal to file a formal notice and blocked finalization of the transaction until its review was completed, arguing that the agreement posed potential "risks to national security."[150] Increased scrutiny by CFIUS also likely discouraged some companies from even considering investments that they otherwise would have pursued. Overall, CFIUS's "more aggressive approach to reviewing investments by China" contributed to the significant decrease in inbound FDI from China since 2017.[151]

As of February 2024, the Department of Commerce had announced $1.5 billion in grants to chipmaker GlobalFoundries to upgrade and expand its chipmaking facilities. It also made two smaller production grants to Microchip Technology Inc. and BAE Systems and has

[146] "China Is Quietly Reducing Its Reliance on Foreign Chip Technology," 2024.

[147] Jonathan O'Callaghan, "Who's Making Chips for AI? Chinese Manufacturers Lag Behind US Tech Giants," *Nature*, May 3, 2024.

[148] Saif M. Khan, "Maintaining the AI Chip Competitive Advantage of the United States and Its Allies," issue brief, Center for Security and Emerging Technology, December 2019, p. 4.

[149] Luuk Klein, "CFIUS Interventions Focus on Semiconductors, Financial Services," Ion Analytics, September 27, 2022.

[150] Damely Perez and Kimberly Shi, "Back to the Future: The Committee on Foreign Investment in the United States (CFIUS) Scuppers Another Semiconductor Transaction," Clifford Chance, January 11, 2022.

[151] Brian J. Egan, Michael E. Leiter, and Tatiana O. Sullivan, "'Small Yard and High Fence': US National Security Restrictions Will Further Impact US-China Trade and Investment Activity in 2024," *Skadden's 2024 Insights*, Skadden, Arps, Slate, Meagher & Flom LLP and Affiliates, December 13, 2023.

announced additional impending awards to other major chip companies.[152] The Biden administration estimates that private companies have made $235 billion in commitments to invest in manufacturing semiconductors and electronics since the passage of the CHIPS and Science Act. U.S. chipmakers have reportedly encountered issues in their plans to expand domestic production, however, leading to longer construction times than initially anticipated.[153]

Conclusion

The U.S. government has enacted a variety of policies to protect U.S. economic, technological, and national security interests. The United States' principal objectives have centered on the control of technologies that could have applications for both commercial and military purposes and the diversification of supply chains to reduce U.S. dependence on Chinese suppliers. Measures to control Chinese investments in key technologies appear to have been successful. Chinese investments have dwindled after the augmentation of CFIUS oversight through the passage of FIRRMA. U.S. efforts to diversify supply chains in critical minerals—such as polysilicon, lithium, graphite, and rare earths—have had mixed results, owing to the different nature of the goods. In the case of polysilicon, lithium, and graphite, nurturing U.S. suppliers is likely to result in reduced dependence on Chinese suppliers over time. However, the United States may need more time to diversify its supply of rare earths because of the cost and difficulty of refining.

U.S. measures to control Chinese access to key technologies, such as advanced semiconductors, appear to have succeeded. Authorities have employed a variety of measures, but the most significant appear to be export controls and restrictions on Chinese investments in U.S. technology companies. Other measures, such as tariffs, appear to have been less effective in achieving the goals of supply diversification and discouraging Chinese acquisition of key technologies (Table 3.6). The Entity List had a significant short-term effect on Huawei, but the

TABLE 3.6
Assessment of U.S. Economic Policies to Defend U.S. Interests

Sub-Objective	Assessment
Reduce excessive dependence on Chinese suppliers	Mixed
Control transfer of key technologies	Success

SOURCES: Contains information from Blinken, 2022, and Yellen, 2023.

[152] U.S. Department of Commerce, "Biden-Harris Administration Announces CHIPS Preliminary Terms with Microchip Technology to Strengthen Supply Chain Resilience for America's Automotive, Defense, and Aerospace Industries," January 4, 2024; Ana Swanson, "Biden Administration Chooses Military Supplier for First Chips Act Grant," *New York Times,* December 11, 2023b; Don Clark and Ana Swanson, "Plans to Expand U.S. Chip Manufacturing Are Running into Obstacles," *New York Times,* February 19, 2024.

[153] Clark and Swanson, 2024.

company is developing various work-arounds. In general, companies that are placed on the Entity List have often been able to find alternative avenues to import at least small amounts of inputs by going through third-party shell companies. However, the list has been effective when substantial, ongoing imports of components are needed to keep operations flowing.

CHAPTER 4

China's Response

In this chapter, we examine China's response to U.S. economic measures pursued between 2017 and 2024. Before we begin, however, we provide additional context by briefly reviewing China's development goals since the 2000s. The intensification of economic competition between China and the United States during this period is due, in part, to the maturation of China's economy and the unsustainability of its reliance on low-cost labor and capital to power its growth. China's decision to become the dominant supplier of a variety of emerging technologies and compete globally in markets for goods featuring these technologies has unavoidably resulted in more-direct economic competition with the United States in particular and the developed West more broadly.

Following our review of China's development goals, we consider Chinese reactions to the economic policies that the United States has adopted under Presidents Trump and Biden. We review the Chinese government's formal responses in both the near and long terms. As we show, Chinese officials have responded with tit-for-tat retaliatory tariffs and trade restrictions while calling for talks to ease tensions and head off a potential escalation of the trade war. Conflicts over trade have also accelerated China's investment in domestic technologies to substitute for goods previously imported from Western countries, thereby making China more self-reliant and reducing its technological dependence on the West. We also examine commentary by Chinese analysts to develop a sense of the Chinese government's perspective on recent U.S. economic measures. Despite acknowledging that these measures have damaged China's economic and technological development, Chinese commentators claim that the United States has suffered more than China and that the measures have largely proven self-defeating.

China's Development Goals

Beginning in the early 2000s, Beijing faced increasing external costs of rapid growth in the form of rampant corruption, worsening inequality, and severe environmental pollution. Not long after, China confronted a more challenging security environment because the United States adopted a more adversarial approach in the face of more belligerent rhetoric and expanding military capabilities on the part of China. China responded to these pressures with a variety of policies, including efforts to expand access to markets in the developing world, improve the competitiveness of its advanced manufacturing sectors, and stabilize ties with the United States.

The evolution of China's economy follows a well-established precedent set by other industrializing states. In the first decades of rapid growth following the shift toward market-friendly reforms in the 1980s, China relied on the mobilization of domestic capital and rural labor. This approach turbocharged China's economy: Annual GDP growth averaged 9.4 percent from 1979 to 2004.[1] The rapid growth lifted 400 million Chinese out of poverty, and per capita incomes increased tenfold from $158 in 1978 to $1,508 in 2004.[2] However, China's ability to squeeze growth from such methods inevitably diminished over time. The number of China's workers started to decline in 2012.[3] Even before then, beginning around the early 2000s, rising wages rendered an economic strategy based on low-cost labor unsustainable.[4] Investment-driven growth reached its limits as well. To maintain high growth rates, the country accumulated massive amounts of domestic debt and wasted resources on unnecessary construction projects and the maintenance of an inefficient state sector.[5] From 2019 to 2021, China's officially reported annual GDP growth averaged between 6 and 7 percent, and growth rates have declined further in response to the shocks from the COVID-19 pandemic.[6] The slowdown became particularly pronounced in 2022 and 2023 when, according to official figures, the economy grew at 3 percent and 5.2 percent, respectively, owing in part to a crisis in the property market.[7]

The compounding effects of high debt, low productivity, and demographic aging are almost certain to result in much slower long-term rates of growth.[8] This slowdown in growth is occurring as China continues to grapple with the legacy costs of its past growth policies, including wide income inequalities, severe pollution, an underdeveloped social safety net,

[1] World Bank, "DataBank: World Development Indicators, China, GDP Growth (Annual %)," database, undated-a.

[2] World Bank, "GDP per Capita (Current US$)—China," interactive graph, World Bank national accounts data and Organisation for Economic Co-Operation and Development national accounts data files, undated-c.

[3] National Bureau of Statistics of China, 2023.

[4] Tom Orlik, "Rising Wages Pose Dilemma for China," *Wall Street Journal*, May 17, 2013.

[5] Lucy Hornby, "China Locked into Investment-Led Growth by GDP Targets," *Financial Times*, July 22, 2019.

[6] World Bank, "GDP Growth (Annual %)—China," interactive graph, World Bank national accounts data and Organisation for Economic Co-Operation and Development national accounts data files, undated-b. Chen et al. argue that China's national statistical office overestimated China's nominal rate of growth in GDP by 1.8 percentage points annually between 2010 and 2016 (Wei Chen, Xilu Chen, Chang-Tai Hsieh, and Zheng Song, *A Forensic Examination of China's National Accounts*, National Bureau of Economic Research, Working Paper 25754, April 2019, rev. December 2019). There is no sign that overestimation has stopped.

[7] Keith Bradsher, "China's Economy Grew Last Year, but Strains Lurk Behind the Numbers," *New York Times*, January 16, 2024.

[8] Derek Scissors, *A Stagnant China in 2040, Briefly*, American Enterprise Institute, March 16, 2020.

and rampant corruption.[9] Some experts project that China's GDP growth rate could slow to 2 to 3 percent on average through the mid-century.[10] In light of the impending declines in China's labor force, real growth rates could be even slower.

Chinese leaders have recognized the need to refine the country's development strategy. The Chinese Communist Party leadership under Hu Jintao attempted to rebalance the economy toward a more "scientific" mode of balanced, sustainable growth, albeit with little success.[11] His successor Xi Jinping has pursued a centralized strategy to maintain economic growth and move China into sectors characterized by higher technological levels. At the 2013 Third Plenum of the 18th Party Congress, Xi announced a major, multi-sector reform initiative to establish fairer rule of law, strengthen economic competitiveness, curb corruption, redress environmental damage, revamp state industries, expand social welfare benefits, encourage technological innovation, and bolster market institutions as part of a broader effort to revamp the economy and shift to a more efficient and more balanced mode of growth. Xi also announced policies designed to dominate global markets and key high-technology sectors, although, in both cases, implementation of all the planned reforms has been poor.[12] These economic initiatives carry important implications for the U.S. economy and have, in part, motivated U.S. policymakers to carry out retaliatory and protective measures, including the substantial increases in tariffs in 2018 and 2019.

A Government-Led Chinese Model

China promotes its political security, in part, by encouraging more countries to adopt growth policies that mimic its own. In a 2019 article published in *Seeking Truth*, President Xi predicted that the Chinese growth model would have increasing influence around the world.[13] In a separate speech, he stated that countries can best ensure their economic security by relying on "both the invisible hand and the visible hand."[14] He explained that China's approach combines the market's ability to allocate resources efficiently with a strong role for the state

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9 Andrea Boltho and Maria Weber, "Did China Follow the East Asian Development Model?" *European Journal of Comparative Economics*, Vol. 6, No. 2, Autumn 2009.

10 Roland Rajah and Alyssa Long, "Revising Down the Rise of China," Lowy Institute, March 14, 2022.

11 Joseph Fewsmith, "Promoting the Scientific Development Concept," *China Leadership Monitor*, No. 11, July 30, 2004.

12 "Communiqué of the Third Plenary Session of the 18th Central Committee of the CPC" ["中国共产党十八届三中全会公报"], China.org.cn, January 16, 2014.

13 Xi Jinping, "Several Issues on Adhering to and Developing Socialism with Chinese Characteristics" ["关于坚持和发展中国特色主义的几个问题"], *Seeking Truth* [求是], March 31, 2019.

14 Xi Jinping, "The 'Invisible Hand' and the 'Visible Hand,'" *Qiushi Journal*, Communist Party of China Central Committee Bimonthly, May 26, 2014.

in controlling key sectors, ensuring equitable social and economic outcomes, stabilizing markets, and solving large-scale problems.[15]

The logic of China's growth model is that development-oriented policies create more jobs and reduce poverty, thereby preventing social unrest and ensuring state security. This logic justifies development-first policies that prioritize infrastructure and logistics-related projects over strengthening governance and political institutions. When countries adopt policies that reflect Beijing's preference, they legitimize China's growth model, which, in turn, may bolster the Chinese Communist Party's legitimacy and sense of security.[16] China's advocacy of this approach is not disinterested; the development-first policy opens new markets for Chinese companies and products.

China's economic model, led by an authoritarian state, is less transparent and, thus, more attractive to other authoritarian regimes. Formal law and formal dispute settlement play reduced roles and are displaced by memoranda of understanding and informal state-to-state and private negotiations.[17] Several Chinese initiatives illustrate this approach, especially BRI. Beijing uses BRI to develop new markets for Chinese products through a trade, investment, and infrastructure network that is governed by a combination of contracts and treaties, backed by bilaterally negotiated dispute resolution mechanisms. These mechanisms encourage economic integration with and dependence on Beijing.[18] This approach contrasts with the one favored by the United States and its friends and allies, which support transparent rules and laws and formal dispute settlement mechanisms that apply to all.

Initiatives to Dominate Foreign Markets

Beijing's announcement of its initiative to set up the Asian Infrastructure Investment Bank in 2013 garnered considerable international attention, but establishing the bank represented just one in a series of large-scale economic initiatives that China has pursued in recent years.[19] In 2013, Xi announced BRI during a state visit to Kazakhstan and pledged up to $2 trillion to help partner countries across Eurasia, Africa, and Latin America upgrade their infrastructure and expand trade and investment ties with China.[20] That year, Beijing also formed a

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[15] Xi Jinping, "Working Together to Forge a New Partnership of Win-Win Cooperation and Create a Community of Shared Future for Mankind," Ministry of Foreign Affairs, The People's Republic of China, September 29, 2015.

[16] Ilaria Carrozza, "China's Multilateral Diplomacy in Africa: Constructing the Security-Development Nexus," in Daniel Johanson, Jie Li, and Tsunghan Wu, eds., *New Perspectives on China's Relations with the World: National, Transnational and International*, E-International Relations, 2019.

[17] Tom Ginsburg, "Authoritarian International Law?" *American Journal of International Law*, Vol. 114, No. 2, April 2020.

[18] Nadège Rolland, *China's Eurasian Century? Political and Strategic Implications of the Belt and Road Initiative*, National Bureau of Asian Research, 2017.

[19] "An Asian Infrastructure Bank: Only Connect," *The Economist*, October 4, 2013.

[20] "Chronology of One Belt, One Road," Xinhua, March 28, 2015.

second international bank, the New Development Bank, based on the Brazil, Russia, India, China, and South Africa (BRICS) group to expand cooperation among emerging economies and reduce their dependence on Western financial institutions and the U.S. dollar.[21] The New Development Bank's operations have been relatively modest, with loan activity severely curtailed following Russia's invasion of Ukraine in 2022.[22] China joined the Regional Comprehensive Economic Partnership, created by the Association of Southeast Asian Nations, in hopes of shaping the regional trade agreement to accord with China's preferences and limit U.S. involvement.[23] It also sought to directly counter U.S. efforts to promote the United States' proposed regional trade initiative, the TPP, which Washington ultimately abandoned.[24] These initiatives aimed to establish a leading role for China in emerging and global markets and threatened to displace U.S. leadership and influence in the Asia-Pacific region.

Another series of initiatives aimed to establish China's leadership in sectors featuring high technology and emerging industrial sectors. The policy "Made in China 2025" is representative of this approach. Launched in 2015, "Made in China 2025" involves massive subsidies for R&D and investments of hundreds of billions of dollars. It aims to turn China into a hub for high-tech manufacturing. Among its objectives, the plan outlines ambitions to increase the domestic share of suppliers for core, high-tech components and materials to 70 percent by 2025.[25] Key sectors highlighted by the government plan include AI, 5G networks, biotechnology, and EVs. To meet these objectives, in addition to the subsidies provided to domestic companies, the Chinese government has pushed Chinese companies to purchase domestic, high-tech products and software rather than imported ones and has provided other forms of assistance.[26]

In the 2000s, China's rising share of global manufacturing eventually led to overcapacity. To keep growth going, the Chinese government provided incentives to shift investment away from manufacturing. This move contributed to a shift in investment to real estate. By the 2010s, construction and other sectors linked to real estate accounted for nearly 30 percent of the country's economic activity, but the relentless pace of residential construction proved unsustainable. Fueled by cheap credit, the real estate market experienced a glut, a phenom-

[21] "Brics Countries Launch New Development Bank in Shanghai," BBC, July 21, 2015.

[22] Alexander Saeedy and Lingling Wei, "A Bank China Built to Challenge the Dollar Now Needs the Dollar," *Wall Street Journal*, June 16, 2023.

[23] Michael D. Sutherland, *Regional Comprehensive Economic Partnership (RCEP)*, Congressional Research Service, IF11891, October 17, 2022.

[24] David Groten, "China's Approach to Regional Free Trade Frameworks in the Asia Pacific: RCEP as a Prime Example of Economic Diplomacy?" *Sicherheit Und Frieden (S+F)* [*Security and Peace*], Vol. 35, No. 3, 2017.

[25] "China to Invest Big in 'Made in China 2025' Strategy," *China Daily*, October 12, 2017.

[26] Jost Wübbeke, Mirjam Meissner, Max J. Zenglein, Jaqueline Ives, and Björn Conrad, *Made in China 2025: The Making of a High-Tech Superpower and Consequences for Industrial Countries*, Mercator Institute for China Studies, December 2016.

enon vividly captured by the specter of empty "ghost cities."[27] Beijing finally directed a reduction in lending to real estate, which caused severe economic distress among major developers, such as Country Garden Corporation, China's largest developer.[28] The slowdown in the real estate sector has aggravated problems of unemployment and threatened the state's goals for economic growth.

Authorities have now directed more lending to export industries, especially in green technologies. As a result, China's share of global markets for lithium-ion batteries, solar panels, and EVs increased sharply from 2019 to 2023.[29] Experts warn that China's efforts to dominate world green energy export markets could threaten the efforts of the United States and European countries to develop their own green industries.[30] However, green industry alone is unlikely to displace growth that had been driven by real estate because the value of green technology industries is a fraction of that associated with the real estate and construction industries.

These initiatives to influence economic policies in developing countries and dominate advanced technological sectors and global export markets pose a challenge to the U.S. and other Western economies. High-tech sectors—such as aviation, semiconductors, and information technology—remain critical to many Western economies, including Japan, South Korea, and the United States. As the United States faces the prospect of competition with China in these sectors, China's economic policies—such as subsidized production, overcapacity, and intellectual property theft—have fueled trade tensions and are a major driver of U.S. economic policies directed at China.

Official Response: Manage Tensions, Respond Tit for Tat

China's response to U.S. economic policy measures cannot be fully separated from its overall approach to relations with the United States. Chinese leaders view China's relationship with the United States as increasingly difficult. Chinese analysts and officials have focused on the changes in China's development goals as one reason why relations with the United States have worsened.[31]

In 2010, researchers in China's Ministry of Commerce concluded that China's future development depends, in part, on its ability to exercise international influence commensurate

[27] "What This $100 Billion Ghost City Says About China's Real Estate Crisis," video, *Wall Street Journal*, September 8, 2023.

[28] Henry Hoyle and Sonali Jain-Chandra, "China's Real Estate Sector: Managing the Medium-Term Slowdown," International Monetary Fund, February 2, 2024.

[29] Peter Tirschwell, "China's Export Dominance Is Getting Stronger, Not Weaker," *Nikkei Asia*, December 6, 2023.

[30] Niels Graham, "China's Manufacturing Overcapacity Threatens Global Green Goods Trade," Atlantic Council, December 11, 2023.

[31] Timothy R. Heath, "China's Evolving Approach to Economic Diplomacy," *Asia Policy*, No. 22, July 2016.

with its status as one of the world's largest economies. The report outlines a vision of China as a leader that shapes the terms of world trade and investment rather than merely a follower of established rules. It uses the term "strong trading power" to describe this vision, a term that senior leaders have also adopted.[32] In December 2014, for example, Xi Jinping called for China to "take part in the formulation of international economic and trade rules."[33] Major economic and geostrategic initiatives to deepen Asia's economic integration through BRI, the Asian Infrastructure Investment Bank, and other initiatives provide just some examples of Beijing's implementation of policies aimed at restructuring the economy of Asia to better support China's economic desires. These sources acknowledge that the United States would likely view China's more assertive approach as threatening and might try to adopt more confrontational measures.[34]

Despite the increase in tensions, Chinese authorities reject the notion that a trade war or armed conflict with the United States is inevitable. China continues to advocate for a balance between actions to defend its interests and actions to stabilize U.S.-China ties. In a typical formulation, Foreign Minister Wang Yi stated in 2019 that "competition is normal" but criticized the idea that the two countries were headed toward confrontation.[35] Similarly, a 2019 foreign policy white paper affirmed China's right to defend its interests yet called on the United States to adopt "cooperation" as the "only correct choice."[36] The combination of these two competing goals—one in which China defends its interests and another in which it encourages cooperation to stabilize relations—underpins China's pursuit of institutionalized ties with the United States, by which the two countries can manage their differences in a nonhostile manner.[37]

In short, Chinese leaders view the increase in tensions with the United States, in part, as an unavoidable consequence of China's rise as a great power. Managing the tensions requires a delicate balance between actions to defend China's interests and actions to stabilize ties with the United States. This imperative can be seen in China's response to such U.S. economic measures as tariffs and restrictions on technology exports. At the same time, Chinese officials advocate talks to reduce bilateral strains.

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[32] Ministry of Commerce Research Institute (China) Topic Group, "Strategic Readjustment and Structural Innovation in China's Foreign Trade Policy in the Post-Crisis Era" ["后危机时代我国对外贸易的战略性调整"], *International Trade* [国际贸易], 2010, pp. 4–10.

[33] Timothy Heath, "Xi's Bold Foreign Policy Agenda: Beijing's Pursuit of Global Influence and the Growing Risk of Sino-U.S. Rivalry," *China Brief*, Vol. 15, No. 6, March 19, 2015.

[34] Rolland, 2017.

[35] "China, U.S. Stand to Gain from Cooperation, Lose from Confrontation: FM," Xinhua, March 8, 2019.

[36] State Council Information Office of the People's Republic of China, *China and the World in the New Era*, white paper, September 2019.

[37] State Council Information Office of the People's Republic of China, 2019.

China's Counter to U.S. Policies

China has responded to U.S. economic sanctions and other measures in several ways. In the near term, Chinese officials have directed tit-for-tat responses, such as increasing tariffs on U.S. exports and imposing restrictions on trade with the United States. However, the trade war has added further impetus to the country's long-standing interest in technological self-reliance and encouraged China to expand its trade relations with other developing countries.

China's government announced its first set of retaliatory tariffs in April 2018, just weeks after the U.S. government increased tariffs on steel and aluminum products. After the U.S. government announced a 25-percentage point increase in tariffs on $50 billion of Chinese imports, Beijing retaliated with its own 25-percentage point increase in tariffs on $50 billion worth of U.S. goods. A few months later, the U.S. government announced a list of imported goods worth $200 billion that were subject to increases in tariffs ranging from 5 to 10 percentage points, to which China retaliated with its own increases in tariffs of about 5 to 10 percentage points on $60 billion worth of U.S. exports to China.[38] In response to tariffs announced by the United States in June 2018, a Ministry of Commerce spokesperson accused the United States of "los[ing] its rationality" and declared that China would "resolutely strike back" with "comprehensive measures."[39] China also retaliated with its own restrictions on foreign ownership of Chinese firms; the government adopted an antitrust regime to block all mergers and acquisitions in China involving U.S. technology firms.[40]

Throughout the exchange of increases in tariffs, Chinese officials maintained a defiant stance yet expressed a willingness to negotiate. For example, in May 2019, a Foreign Ministry spokesperson stated that China would "never surrender to external pressure" and blamed the United States for "abandon[ing]" a consensus on economic and trade issues. At the same time, the spokesperson expressed the hope that the two countries could "meet each other halfway and achieve a mutually beneficial and win-win agreement on the basis of mutual respect, equal treatment and commitment."[41] Similarly, in 2021, Chinese authorities continued to express a willingness to resume talks while criticizing U.S. behavior. A spokesperson urged the United States to "'listen carefully to rational voices within the country and work with China to create a favorable atmosphere for the healthy development of bilateral economic and trade cooperation.'"[42] The following year, Chinese authorities continued to lambast U.S. trade sanctions when the Ministry of Foreign Affairs posted a rebuttal to Sec-

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[38] Andrew Mullen, "US-China Trade War Timeline: Key Dates and Events Since July 2018," *South China Morning Post*, August 29, 2021.

[39] "China MOC Spokesperson Makes Remarks on White House Trade Statement," Xinhua, June 19, 2018.

[40] Gregory C. Allen, *China's New Strategy for Waging the Microchip Tech War*, Center for Strategic and International Studies, May 3, 2023.

[41] "China Will Never Surrender to External Pressure: FM Spokesperson," Xinhua, May 15, 2019.

[42] "Essence of China-U.S. Economic, Trade Relations Mutually Beneficial: Spokesperson," Xinhua, January 15, 2021.

retary of State Blinken's comments on China. The document carried a searing indictment of U.S. policy, leveling accusations of self-dealing, hypocrisy, and incessant warmongering. Citing U.S. export restrictions regarding technologies, the rebuttal accused Washington of "violat[ing] . . . the principle of fair competition and market economy and international trad[e] rules." It claimed the United States sought to "hamstring competitive Chinese hi-tech companies under all kinds of trumped-up charges." It also stated the United States had "waged a massive trade war on China" and called on Washington to resume negotiations.[43] In 2023, Xi called for a "mutually beneficial" trade relationship and demanded the United States stop using the reason of national security to "stifl[e] China's technological progress" and "right to development."[44]

Self-Reliance

The trade war has also fueled Beijing's interest in deepening China's technological self-reliance. The roots of this imperative stretch back to the earliest days of Maoist rule, when the Communist government aimed to build a largely autarkic economy in the face of Cold War–era U.S. trade restrictions levied against Communist states.[45] China's government eased off this policy in the early decades of the reform and opening-up era in hopes that foreign investment and trade could expand access to advanced technologies and accelerate the country's development.

As China's rate of economic growth decelerated, however, Beijing became much more interested in moving up the value-added chain and competing in sectors featuring advanced technologies by developing indigenous technologies. In 2015, China's State Council set the goal of becoming a global leader in a variety of emerging and cutting-edge technologies through the "Made in China 2025" program.[46] After the trade war erupted, officials stepped up their calls for China to become more self-reliant. On April 10, 2020, Xi gave a speech at the seventh meeting of the Central Financial and Economic Affairs Commission in which he stated, "We must tighten international production chains' dependence on China, forming powerful countermeasures and deterrent capabilities based on artificially cutting off supply to foreigners."[47]

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[43] Ministry of Foreign Affairs of the People's Republic of China, "Reality Check: Falsehoods in US Perceptions of China," June 19, 2022.

[44] "Xi, Biden Hold Historic Summit, Charting Course for Improving Bilateral Ties," Xinhua, November 17, 2023.

[45] Mary Hui, "China Was "De-Risking" Long Before the Term Caught On in the West," *Quartz*, September 14, 2023.

[46] Institute for Security and Development Policy, *Made in China 2025*, June 2018.

[47] Xi Jinping, "Major Issues Concerning China's Strategies for Mid- to Long-Term Economic and Social Development" ["国家中长期经济社会发展战略若干重大问题"], *Seeking Truth* [求是], October 31, 2020.

For Chinese leaders, self-reliance involves issues of technology, manufacturing, energy, finance, and food security. In 2016, Xi said, "the fact that core technology is controlled by others is our greatest hidden danger."[48] And in a September 2018 speech, Xi emphasized "the importance of self-reliance in food security, the real economy and manufacturing." He admitted it is "becoming increasingly difficult to gain access to leading technologies and key technologies internationally" and said that this would make China "ultimately rely on itself."[49] The 14th Five-Year Plan outlines measures to "achieve security and controllability in critical areas such as important industries, infrastructure, strategic resources, and major [science and technology] fields." The Plan includes sections dedicated to food, energy, and financial security.[50]

China's government has directed measures to lessen the country's vulnerability, including stockpiling such key high-tech products as advanced semiconductors and semiconductor manufacturing equipment. China's government has also continued to massively subsidize the growth of China's semiconductor industry with a $21 billion National Integrated Circuit Industry Fund, launched in 2014 and renewed in 2019 with additional funding.[51]

Chinese Commentators Admit Damage but Argue the United States Is Suffering Worse

Chinese commentators have supported official government policy with several lines of criticism. Chinese media admit that recent U.S. trade policies vis-à-vis China have inflicted harm but counter that the United States has suffered more. Commentators also argue that the trade war has been a failure and that the U.S. measures are damaging the United States' reputation with the world and harming the global economy.

Some Chinese commentators admit that the U.S. measures have impaired China's economic and technological development. Chinese scholars and experts have held seminars to evaluate the results of the trade war. [52] One 2022 commentary in the *Global Times* acknowledged that the sanctions had "suppress[ed] China's technological and economic development;" however, it claimed that the United States had also suffered.[53] He Weiwen, a senior fellow at the Center

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48 "Xi's Remarks on Cyber Security, Informatization Published," Xinhua, April 27, 2016.

49 Xu Wei, "Xi Stresses Nation's Self-Reliance," *China Daily*, September 27, 2018.

50 Etcetera Language Group, Inc., trans., *Outline of the People's Republic of China 14th Five-Year Plan for National Economic and Social Development and Long-Range Objectives for 2035* [中华人民共和国国民经济和社会发展第十四个五年规划和 2035 年远景目标纲要], Center for Security and Emerging Technology, May 12, 2021.

51 Allen, 2023.

52 "Changing China-U.S. Trade Ties Require New Global Outlook," *Global Times*, April 24, 2023.

53 Huo Jianguo, "US Trade Policy Toward China Should Come Back to Rational in 2023," *Global Times*, December 29, 2022.

for China and Globalization, stated in a 2023 article that U.S. tariffs on Chinese merchandise "did harm to both sides, but companies in the US sustained more damage."[54] Another *Global Times* commentator claimed that the U.S. measures were "restricting cooperation with China in the field of science and technology" and had "backfir[ed] against the US economy itself."[55]

Commentators often claim the tariff measures have "failed miserably," citing evidence that the U.S. trade deficit expanded after adopting the tariffs.[56] Another typical commentary stated that the trade war has been a "complete failure."[57] Accusations that the United States should bear the blame for a sluggish global economy also appear in Chinese commentary.[58] A 2023 article stated that the United States had brought "great uncertainties to the global trading system" and that it had "sabotage[ed] a much-needed driving force for world economic recovery from the COVID-19 pandemic." [59]

Another line of argument claims that Washington harbors an ulterior motive. Commentators accuse the United States of using the measures to suppress China and sustain U.S. hegemony. A 2022 commentary stated that U.S. "trade protectionism" and "abuse of the concept of national security" revealed that Washington intended to do "whatever it takes to maintain its hegemony." [60] Famed Chinese international relations expert Shi Yinhong, a professor at People's University, accused the United States of "form[ing] cliques" with its allies to "contain China in terms of high tech, and restructure industrial and supply chains."[61]

In academic and technical journals, economists and scholars have expressed concern about the ways the U.S. government has succeeded in constraining China's access to advanced technology. They note that Western countries have developed "choke points" to suppress China's technological development.[62] These scholars have also suggested ways to counter U.S. economic policies toward China. In 2018, a series of articles in *Science and Technology Daily* discussed key and core technologies controlled by the United States and its allies. The articles cited high-end electronic components and specialized steel as among the

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[54] "China Urges US to Drop All Additional Tariffs as Washington's Trade War Drags On After 5 Years," *Global Times*, March 23, 2023.

[55] Jianguo, 2022.

[56] Wang Cong, "Opinion: US' Trade War with China Is a Lost Cause; Biden Has No Choice but to End It," *Global Times*, May 13, 2022.

[57] "Reflection on Trade War Failure a Much-Needed Step for US," *Global Times*, May 30, 2023.

[58] Yifan Xu, "'De-Risking' Another Term for Decoupling," *China Daily*, July 11, 2023.

[59] "Trade with China Not a Buffet Where US Can Pick and Choose," *Global Times*, March 22, 2023.

[60] Jianguo, 2022.

[61] "Changing China-U.S. Trade Ties Require New Global Outlook," 2023.

[62] Ben Murphy, *Chokepoints: China's Self-Identified Strategic Technology Import Dependencies*, Center for Security and Emerging Technology, May 2022.

most-acute "chokepoints" facing the country.[63] In the series, the authors pointed out how, in some cases, Western countries have "monopolized" the strategic technology in question. In other cases, China lacks market share in critical technologies. The authors expressed pessimism about the consequences for China's development if the United States imposes trade restrictions or sanctions that cut off access to vital technologies.[64]

Conclusion

U.S. trade and economic measures against China have unfolded against a backdrop of deteriorating U.S.-China relations. China's government has responded to U.S. trade and other economic sanctions by retaliating in a tit-for-tat manner through the imposition of tariffs and restrictions on the ability of U.S. technology firms to acquire Chinese firms. The escalating trade tensions have fueled concerns in China about the country's vulnerability and added impetus to a long-standing interest in increasing the country's self-reliance on domestic technologies. Authorities have directed measures to increase food security and reduce the country's dependence on foreign technologies. Steps include massive subsidies for the development of indigenous technologies that compete against those of the West and requirements that Chinese companies use Chinese technologies.

Echoing official statements, commentary in Chinese media has generally carried a variety of sharp criticisms of U.S. policy while affirming China's willingness to negotiate. Lines of argument claim that U.S. economic policies toward China have failed to achieve their stated goals of reducing the U.S. trade deficit and have harmed the global economy and the U.S. economy itself. However, Chinese policies seem to indicate the contrary because the Chinese government appears very concerned about the economic implications of reduced access to U.S. markets and technologies.

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63 Zhang Gailun [张盖伦] and Fu Lili [付丽丽], "ZTE's Chip Problem Gives China Heart Palpitations" ["中兴的' 芯' 病，中国的心病"], *Science and Technology Daily* [科技日报], April 20, 2018.

64 Jiao Yang [娇阳], "Airworthiness Standards: Another Difficult Hurdle for Domestic Aircraft Engines" ["适航标准：国产航发又一道难迈的坎儿"], *Science and Technology Daily* [科技日报], May 11, 2018.

CHAPTER 5

Overall Assessment and Recommendations

In this chapter, we provide our evaluation of the effectiveness of U.S. economic policies regarding China and offer some recommendations on how to improve those policies.

Trade and Investment

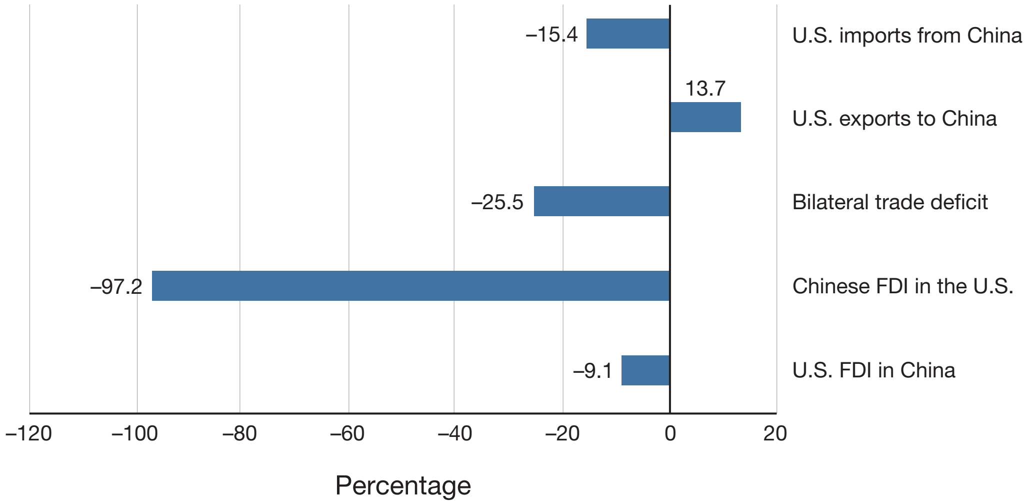
U.S. trade and investment policies pertaining to China have achieved mixed results. Increased tariffs on imports of Chinese goods have been the primary instrument employed by the U.S. government to achieve its goals for fairer trade. The sharp increases in U.S. tariffs on Chinese goods in 2018 resulted in a reduction in imports from China and a sharp decrease in China's share of total U.S. imports between 2017 and 2023. The tariff increases also resulted in a reduction in the bilateral trade deficit, a primary goal of the original decision to impose tariffs (Figure 5.1). Despite the imposition of tariffs on U.S. exports by China, surprisingly, U.S. exports to China increased over this same period.

The decline in U.S. imports from China was especially large between 2022 and 2023, when imports fell 20.3 percent in just one year. All major import categories registered double-digit declines, including machinery and electronics (which have consistently accounted for roughly half of all U.S. imports from China); chemicals, plastics, and leather products; and textiles and footwear. The pattern of these declines suggests that U.S. supply chains are shifting out of China. The tariffs also narrowed the bilateral trade deficit and provided policymakers in the United States a sense of greater balance in the U.S.-China bilateral relationship.

These tariffs, however, were not without cost. The IMF estimates that the tariffs would reduce U.S. GDP by 0.1 percentage points and China's GDP by 0.2 percentage points on an ongoing basis. U.S. manufacturers have suffered losses in export sales to foreign competitors that have been able to undercut U.S. prices because they do not have to pay tariffs on key components. The declines in U.S. manufactured exports led to reductions in manufacturing output and employment because higher sales and employment gains in protected industries failed to offset the negative effects of the tariffs.

FIGURE 5.1

Percent Changes in U.S.-China Trade and Foreign Direct Investment Between 2017 and 2023



SOURCES: Data are from U.S. Census Bureau, undated-a, and Bureau of Economic Analysis, 2024.

Controls on Technology

U.S. economic policies designed to defend U.S. interests consist of export controls, entity lists, controls on foreign investment, and industrial policies. Export controls apply to intellectual property, such as designs or software, as well as physical products. They are the primary means by which the U.S. government prevents the transmission of U.S. technologies to countries that are designated as adversaries or otherwise considered undesirable destinations for U.S. technologies.

According to reactions by the Chinese government, Chinese commentators, and Chinese companies that have relied on recently controlled, imported U.S. products and technologies, the controls have made it more difficult for China to manufacture goods incorporating controlled components. The controls on advanced semiconductors are likely to slow China's ability to develop AI and other technologies.[1] The controls on exports of sophisticated chip manufacturing equipment will make it difficult for China to manufacture its own advanced chips. Some commentators say that, because of these controls, China will be unable to match Western chip manufacturing capabilities for at least ten years. Chinese companies that have been placed on the Entity List by the U.S. Department of Commerce, Huawei in particular, have suffered substantial disruptions to their operations.

[1] Yufeng Xiao, "The Impact of the US-China Trade War on China's Semiconductor Industry," Proceedings of the 2022 2nd International Conference on Financial Management and Economic Transition (FMET 2022), *Advances in Economics, Business, and Management Research*, Vol. 227, 2023.

More-intensive U.S. scrutiny of Chinese FDI in high-tech companies following the passage of FIRRMA in 2018 led to sharp declines in Chinese FDI in the United States. By 2022, Chinese FDI had fallen 97 percent compared with the inflow in 2017. Although China has never been a major source of FDI in the United States, the decline in inflows is striking because FDI from China was not solely dedicated to high-tech companies. For example, Hong Kong's WH Group, which also owns China's largest meat-packer, acquired Smithfield Foods, a U.S. meat processor, in 2013 for $4.7 billion.[2] Thus, FIRRMA is not the only reason for the decline in Chinese FDI in the United States. Because inflows from China are small compared with inflows from other major U.S. economic partners, the economic costs of the decline in Chinese FDI are tiny.

Initial responses from industry to the Biden administration's programs to subsidize investments in manufacturing, mining, and R&D in the United States have been positive. Several U.S. and foreign companies have announced investments in new semiconductor plants in the United States. Additionally, several companies have announced plans to expand or open mines or production facilities for lithium, graphite, and rare earths; a few are planning on investing in refining operations as well.

A recurring problem with industrial policies is the creation of excess capacity and the failure of recipients to produce products at cost-competitive levels. Cost overruns on investment projects are frequent. In some industries, such as solar cells and solar panels, it is doubtful that U.S. manufacturers will be able to compete with Chinese companies. Consequently, the ultimate success of these policies is still unknown.

Conclusion

In sum, U.S. economic policies vis-à-vis China have had mixed success (Table 5.1). They have made progress in promoting fairer trade and provided a higher degree of success in defending U.S. economic-related interests. U.S. tariffs have succeeded in reducing imports and the bilateral trade deficit with China. The United States has also made some progress in shaping new international trade rules and norms in the Indo-Pacific region and, in some areas, appears to be on the road to diversifying its supply chains away from an excessive dependence on China. Industrial policies have resulted in several investments in new semiconductor plants and in mines and refineries for critical minerals. The policies have had some success in reducing Chinese access to sensitive U.S. technologies through licit means. However, these economic policies have had very little success in ensuring fairer treatment for U.S. firms in China and even less in persuading the Chinese government to reduce its subsidies and other anticompetitive state assistance to exporters. The policies have not been free of costs. The United States has paid a price through slower rates of economic growth, lower exports, and losses in manufacturing jobs and output. Despite these costs, the eco-

[2] Shruti Date Singh, "Shuanghui Agrees to Acquire Smithfield Foods for $4.72b," Bloomberg, May 29, 2013.

TABLE 5.1
Effectiveness of U.S. Economic Policies Toward China

Goal	Sub-Objectives	Assessed Progress
Promote fair trade	Reduce imports from China	Success
	Reduce bilateral trade deficit	Success
	Increase U.S. exports to China	Mixed
	Diversify supply chains	Mixed
	Shape international trade rules and norms	Mixed
	Convince China to treat U.S. companies equally with Chinese companies	Failure
	Convince China to stop its unfair trade practices	Failure
Defend interests of the United States and its allies	Reduce excessive dependence on Chinese suppliers	Mixed
	Control transfer of key technologies	Success

SOURCES: Contains information from Blinken, 2022, and Yellen, 2023.

nomic policies of both the Trump and Biden administrations have managed to achieve some security and economic goals.

Recommendations

We conclude with recommendations for U.S. economic policy regarding China. We propose refinements to current policies that could achieve better outcomes in terms of the two overarching goals of promoting fair trade and defending U.S. interests. We provide recommendations regarding trade, controls on technology, economic diplomacy, foreign investment, industry, and diversification of supply chains away from China.

Trade

The U.S. government should maintain higher tariffs on imports of goods from China (1) of which China is the dominant supplier and that the Departments of Defense and Commerce consider key technologies and (2) that could undermine U.S. industries considered critical to U.S. economic or national security. In a continued effort to further encourage diversification of its supply chains, the United States should maintain higher tariffs on materials and products of which China is the dominant or sole supplier. The U.S. government should also maintain tariffs on products for which China is seeking to acquire a dominant share of the global market through unfair industrial practices, as in its heavily subsidized EV sector.

To maintain the overall competitiveness of U.S. manufacturing and to benefit U.S. consumers, the U.S. Trade Representative should offer to negotiate reductions of U.S. tariffs on nonsensitive imports of consumer goods and manufacturing inputs in exchange for reductions in Chinese tariffs on U.S. goods. U.S. manufacturers would benefit from being able to purchase materials and components from China at the same prices as their competitors. U.S. consumers would benefit from lower prices on a variety of consumer goods imported from China.

The United States played the leading role in negotiations for the TPP but then abandoned the initiative. The Biden administration has embarked on a new initiative, IPEF. However, this agreement is inadequate to achieve U.S. trade policy goals. The United States faces much higher average tariffs on its exports to Vietnam (9.5 percent) and Malaysia (6.1 percent) than members of the CPTPP, the successor agreement to the TPP. The United States would disproportionately benefit from reductions in the tariffs imposed on U.S. exports if it joined the CPTPP, while Vietnam and Malaysia already enjoy relatively low tariffs on their exports to the United States.

We recommend that the Executive Office of the President reverse current policies and ask the U.S. Trade Representative to enter negotiations to join the CPTPP. Membership in the CPTPP would better achieve U.S. economic policy goals than the IPEF. It would substantially reduce tariffs faced by U.S. exporters, eliminating the advantages U.S. trade competitors, such as Japan and South Korea, enjoy because of duty-free access to the other members of the CPTPP.

Controls on Technology

China has frequently stolen or impinged on the intellectual property of people and organizations in the United States, as well as its allies and partners. Consequently, the United States and its allies and partners have a united view regarding the importance of stopping China from continuing to infringe on the intellectual property of their citizens and organizations.[3]

To address these concerns, the United States should continue to impose export controls on critical technologies. Congress should increase funding for the BIS and other bureaus that are engaged in reviewing critical and emerging technologies to ensure that they have adequate capacity to review proposed items for export controls.

The United States and other governments have become cautious about technological exchanges with China. The imposition of tighter export controls, the termination of technological exchanges, and the addition of Chinese institutions and companies to the Entity List have greatly circumscribed technological interactions with China. The U.S. government should continue to decline official exchanges between U.S. government agencies and their Chinese counterparts involving applied technologies.

Controls on technology exports to China will only be successful if coordinated and implemented with similar policies adopted by U.S. allies and partners. Without the support of

[3] Ursula von der Leyen, "Speech by President von der Leyen on EU-China Relations to the Mercator Institute for China Studies and the European Policy Centre," European Commission, March 30, 2023.

other countries that manufacture competing products, U.S. export controls are toothless. The White House should organize a formal committee to coordinate economic policy toward China across U.S. government agencies.

The U.S. Department of State should ensure that U.S. embassies, especially those located in allied and partner countries, work closely with the China House to coordinate with allies and partners when implementing new export controls or other economic policies vis-à-vis China.

Restrictions on interactions between academics in China and the United States can impair scientific advances. Accordingly, the U.S. government should continue to permit academic collaboration and exchanges with China involving basic research. It should instruct the appropriate officials in the State Department and the Department of Homeland Security to ensure that Chinese academics are able to obtain U.S. visas quickly and easily and, once a visa is granted, are able to enter the United States without interference.[4] The United States should also allow Chinese graduate students and those Chinese students who remain in the United States after graduation to work for U.S. companies under the U.S. government's optional practical training visa program to be able to obtain multiple entry visas so that they can visit friends and families in China while working under this program.

Economic Diplomacy

China's opaque, government-to-government approach to international trade is designed to maximize advantages for China's interests. Such an approach impedes the fair and free flow of goods and services. The State Department should work with allies to highlight the costs of China's approach to its foreign commercial and diplomatic counterparts in countries contemplating entering into these types of agreements. These costs include preferable treatment of imports from China at the cost of foreclosing other sources of supply, favorable treatment of Chinese construction companies for projects, the use of Chinese rather than host country workers on projects, and unfavorable agreements on supplying natural resources to China without regard to their environmental costs.

The investments and opportunities offered by China through BRI and related initiatives may not benefit recipient countries. Poorly structured loan agreements for projects that are not financially viable have saddled recipients with burdensome debt loads. Some of the projects have inflicted considerable damage on the environment. The State Department should work with allies to offer technical advice to countries contemplating taking on loans and investments from China. This advice should focus on ensuring that the proposed project is financially viable, does not increase the country's overall debt burden, is environmentally sound, does not include construction contracts that favor Chinese firms, and primarily employs local rather than Chinese workers.

[4] Fiona Quimbre, Peter Carlyon, Livia Dewaele, and Alexi Drew, *Exploring Research Engagement with China: Opportunities and Challenges*, RAND Corporation, RR-A1839-1, 2022.

Foreign Investment

Aside from increased scrutiny from CFIUS following the passage of FIRRMA, the United States does not seek to bar Chinese FDI in the United States. China also continues to permit U.S. companies to invest in China. To ensure that both countries continue to benefit from these investments, the U.S. government should make it clear through outreach to the Chinese government, companies, and investors that Chinese FDI that does not threaten U.S. national security continues to be welcome. Additionally, the United States should continue to support U.S. FDI in China by pressing Chinese officials to treat U.S. and other foreign companies the same as they treat domestic Chinese companies and ensure these companies are not unfairly prosecuted under Chinese laws. Finally, Congress should increase funding for CFIUS to ensure it has adequate capacity to review proposed investments.

Industry

Industrial policies have a mixed record. Government subsidies for investments and operating costs of favored industries have frequently led to overcapacity and inefficiency. Facilities constructed with subsidies often struggle to compete in the global marketplace because their operating costs are too high. To forestall such outcomes, we recommend that the U.S. government confine future subsidies to industries that have clear implications for U.S. national security and in which China is the predominant supplier. The U.S. government should cease providing subsidies to companies in sectors that are not vital for U.S. national security and in which Chinese firms have a clear comparative advantage, such as solar cells. Additionally, the National Science Foundation and the Departments of Energy and Defense should increase funding on R&D regarding alternatives to critical minerals, technologies that lower consumption of these minerals, and recycling technologies for these minerals. Alternative technologies should reduce the need for new mines and refineries.

Diversification of Supply Chains

The Department of Commerce should set up a unit responsible for strategic planning for critical materials and products that are important for national security. The unit would coordinate policy for the entire supply chain, identifying choke points where investments are needed to ensure that the United States is not reliant on China for the product. DoD's "mine-to-magnet" approach to making strategic investments across multiple stages of the rare earth supply chain is an example of such a comprehensive approach.[5]

The Departments of State and Commerce should work through the Minerals Supply Partnership to coordinate programs with partner countries to "friendshore" parts of critical supply chains.

[5] C. Todd Lopez, "DOD Looks to Establish 'Mine-to-Magnet' Supply Chain for Rare Earth Materials," U.S. Department of Defense, March 11, 2024.

APPENDIX A

Comparison of U.S. and Chinese Statistics on Bilateral Trade and Investment

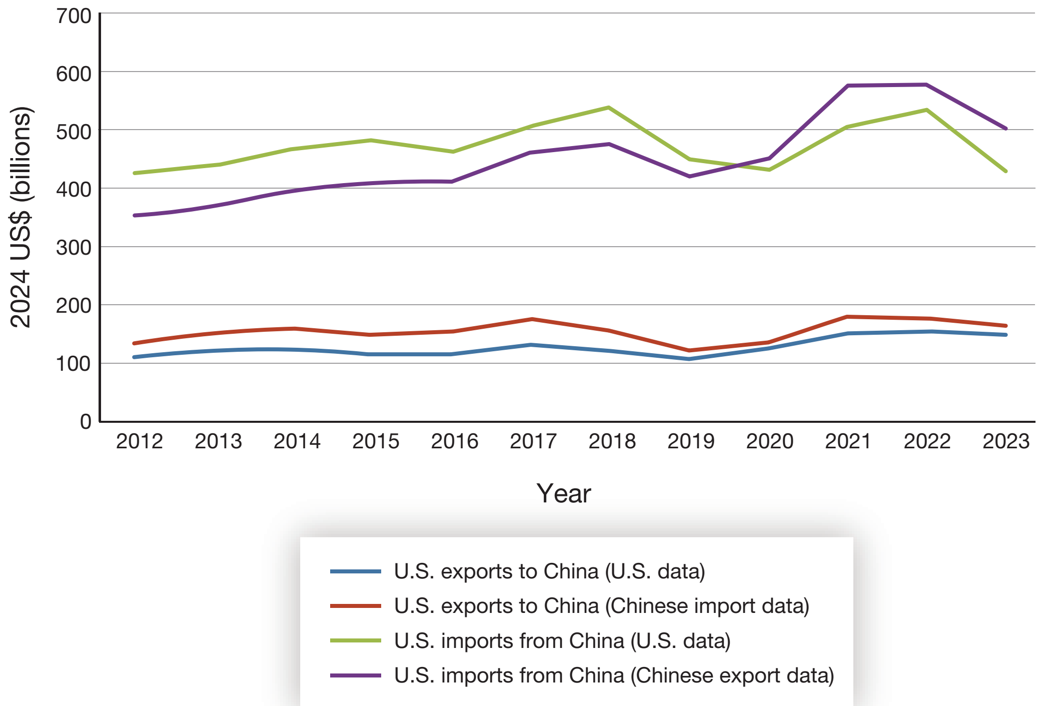
For reasons of clarity and to simplify the analysis, we only use U.S. statistics on trade in the body of this report, even though China publishes mirror statistics. To provide some additional information on trends in trade and investment, we present Chinese statistics below and compare them with U.S. statistics. The statistics from the two countries differ for a variety of reasons. We elucidate some of those differences here.

Statistics on Trade

U.S. statistics on imports from China include costs, insurance, and freight (CIF), whereas the mirror statistics, Chinese exports to the United States, are reported as "free on board" (i.e., they do not include CIF). By the same token, the United States reports U.S. exports to China as free on board, while China reports its imports from the United States as CIF. Consequently, among other differences, the two mirror flows differ by these costs. In addition, the two countries treat exports and imports from and to Hong Kong and Macao somewhat differently. The timing of trade flows can also create differences because U.S. exports are reported when they depart the United States, while China reports imports when they arrive, and vice versa. These differences in timing affect monthly trade statistics and can affect annual trade statistics. There are also differences in attribution to countries of origin. China counts some exports to the United States that the United States attributes to intermediary countries, and vice versa.

Figure A.1 shows these differences between U.S. and Chinese data. Data for the United States come from the U.S. Census Bureau, while data from China are reported by its National Bureau of Statistics. As can be seen, both countries report sharp declines in bilateral trade in 2019, the year following the increases in U.S. tariffs on Chinese goods and Chinese tariffs on U.S. goods. Despite the tariffs, in 2021 and 2022 (during the COVID-19 pandemic), U.S. consumers sharply increased expenditures on goods, including from China. The year 2022 marked an all-time peak for Chinese exports to the United States. In 2023, both countries reported sharp declines in U.S. purchases from China, with U.S. import data reporting a 20.3 percent drop to $427.2 billion, the lowest level of imports from China since 2012.

FIGURE A.1
U.S.-China Trade as Reported by Both Countries



SOURCES: International trade information derived from U.S. Census Bureau, undated-a, and multiple editions of National Bureau of Statistics, *China Statistical Yearbook*, various years (1999–2023).

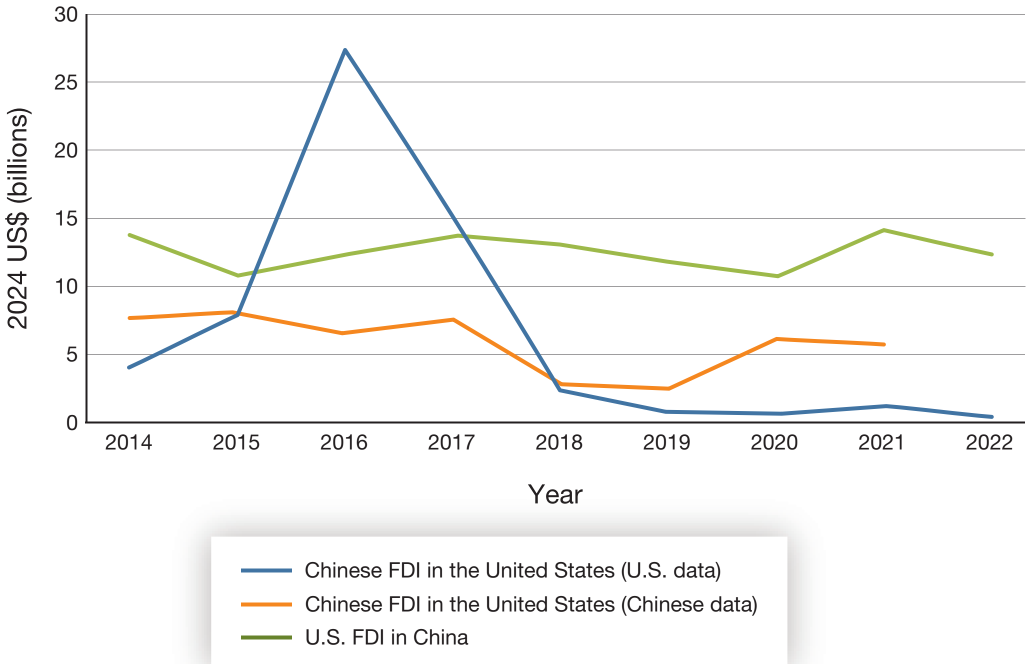
Chinese export data showed a 13.4 percent decline. According to Chinese data, the share of China's total exports that went to the United States peaked at 20.4 percent in 2017; in 2023, the share had fallen to 14.8 percent, a decline of 5.6 percentage points.

Changes in the values and shares of U.S. exports to China have not fallen as sharply as Chinese exports to the United States. According to Chinese data, the share of U.S. exports to China has fallen from 9.5 percent in 2017 to 6.4 percent in 2023, a decline of 3.1 percentage points. According to U.S. data, the share of total U.S. exports going to China fell from 8.4 percent in 2017 to 7.3 percent in 2023, a decline of 1.1 percentage points.

Statistics on Foreign Direct Investment

As with trade statistics, Chinese statistics on FDI differ from those of the United States. As can be seen in Figure A.2, the United States reported much larger inflows between 2015 and 2017 than China reported, followed by much steeper declines. Although both countries reported declines in 2018 and 2019 following the passage of FIRRMA, China reported FDI

FIGURE A.2
Flows of Foreign Direct Investment Between China and the United States



SOURCES: Data are from Bureau of Economic Analysis, 2024, and multiple editions of National Bureau of Statistics, *China Statistical Yearbook*, various years (1999–2023).

flows into the United States of $6.0 billion in 2020 and $5.6 billion in 2021, while the United States reported that inflows from China were only $0.6 billion in 2020 and $1.0 billion in 2021. By either measure, Chinese FDI in the United States has fallen sharply from its 2015 to 2017 levels.

APPENDIX B

Timeline of U.S. Economic Measures Related to Competition with China

Table B.1 summarizes major actions taken since 2000 by either the U.S. Congress, the U.S. President, or a cabinet secretary regarding economic competition with China.

TABLE B.1

Timeline of U.S. Economic Measures Related to Competition with China

Measure	Date Implemented	Objective	Source
U.S.-China Trade Relations Act of 2000	2000	Integrate China into the liberal international order	Public Law 106-286, An Act to Authorize Extension of Nondiscriminatory Treatment (Normal Trade Relations Treatment) to the Peoples Republic of China, and to Establish a Framework for Relations Between the United States and the People's Republic of China, October 10, 2000.
U.S.-China Strategic Economic Dialogue (later, Strategic and Economic Dialogue; Comprehensive Economic Dialogue)	2006–2017	Maintain strong and mutually beneficial U.S.-China economic relations	U.S. Department of the Treasury, "U.S.-China Strategic and Economic Dialogue," webpage, last updated March 9, 2009.
Global financial crisis	2008		
TPP	2008–2017	Enhance U.S. economic posture vis-à-vis China by enforcing U.S. preferred norms and rules	White House, "Statement by the President on the Signing of the Trans-Pacific Partnership," February 3, 2016.
Copenhagen climate compromise	2009	Enhance global cooperation on climate change through the United Nations Framework Convention on Climate Change	Copenhagen Accord of the United Nations Framework Convention on Climate Change, draft decision, December 7–19, 2009.
Cooperation on Climate Change	2011	Enhance cooperation on climate change, clean energy, and the environment	White House, "U.S.-China Cooperation on Climate Change, Clean Energy, and the Environment," fact sheet, January 19, 2011.
Secretary Clinton outlines "pivot to Asia"	2011	Increase U.S. engagement in Asia-Pacific region	Hillary Clinton, "America's Pacific Century," *Foreign Policy*, October 11, 2011.
Joint statement on climate change	2014	Prevent shared challenges to Chinese and U.S. economies	White House, "U.S.-China Joint Announcement on Climate Change," November 12, 2014.
100-Day Plan	2017	Mitigate the bilateral trade deficit by increasing U.S. exports to China	U.S. Department of the Treasury, "Initial Results of the 100-Day Action Plan of the U.S.-China Comprehensive Economic Dialogue," press release, May 11, 2017.

Table B.1—Continued

Measure	Date Implemented	Objective	Source
FIRRMA	2018	Prevent investments in Chinese firms that pose a national security threat	Public Law 115-232, John S. McCain National Defense Authorization Act for Fiscal Year 2019, August 13, 2018.
National Defense Authorization Act for Fiscal Year 2019, section 889	2018	Remove and prevent telecommunications equipment made by Chinese firms from U.S. government systems	Pub. L. 115-232, 2018.
Tariffs targeting China	2018–2019	Reduce imports from and the bilateral trade deficit with China	Schwarzenberg and Hammond, 2019.
Justice Department's "China Initiative"	2018–2022	Prevent theft of intellectual property by Chinese firms	National Security Division, "Information About the Department of Justice's China Initiative and a Compilation of China-Related Prosecutions Since 2018," archived article, U.S. Department of Justice, last updated November 19, 2021.
Justice Department's Huawei indictment	2019	Punish Huawei and its affiliates for evading Iran sanctions	Office of Public Affairs, "Chinese Telecommunications Conglomerate Huawei and Huawei CFO Wanzhou Meng Charged with Financial Fraud," press release, U.S. Department of Justice, January 28, 2019.
China designated as currency manipulator	2019	Mitigate the trade advantage China gains by currency manipulation	U.S. Department of the Treasury, "Treasury Designates China as a Currency Manipulator," press release, August 5, 2019.
Phase One trade agreement	2020	U.S.-China trade agreement	Economic and Trade Agreement Between the Government of the United States and the Government of the People's Republic of China, 2020.
Presidential Proclamation 10043	2020	Prevent Chinese students associated with the People's Liberation Army from accessing U.S. intellectual property	Proclamation 10043, "Suspension of Entry as Nonimmigrants of Certain Students and Researchers from the People's Republic of China," Executive Office of the President, May 29, 2020.
Hong Kong Autonomy Act	2020	Impose sanctions on Hong Kong and Chinese officials	Public Law 116-149, Hong Kong Autonomy Act, July 14, 2020.

Table B.1—Continued

Measure	Date Implemented	Objective	Source
Uyghur Human Rights Policy Act	2020	Impose sanctions on Chinese officials	Public Law 116-145, Uyghur Human Rights Policy Act of 2020, June 17, 2020.
Holding Foreign Companies Accountable Act	2020	Pressure China to allow the Public Company Accounting Oversight Board to conduct inspections of Chinese company audits	Public Law 116-222, Holding Foreign Companies Accountable Act, December 18, 2020.
Executive Orders 13959 and 14032	2020 (expanded 2021)	Ban U.S. investment in "Communist Chinese military companies"	Executive Order 13959, "Addressing the Threat from Securities Investments That Finance Communist Chinese Military Companies," Executive Office of the President, November 12, 2020. Executive Order 14032, "Addressing the Threat from Securities Investments That Finance Certain Companies of the People's Republic of China," Executive Office of the President, June 3, 2021.
Secretary of State determination that China is committing atrocities in Xinjiang	2021	Pressure China regarding human rights abuses	Pompeo, Michael R., "Determination of the Secretary of State on Atrocities in Xinjiang," U.S. Department of State, January 19, 2021.
U.S.-China Joint Glasgow Declaration on Enhancing Climate Action in the 2020s	2021	Advance cooperation on climate change	Office of the Spokesperson, "U.S.-China Joint Glasgow Declaration on Enhancing Climate Action in the 2020s," U.S. Department of State, November 10, 2021.
Secure Equipment Act 2021	2021	Prohibit import of telecoms equipment from Huawei and ZTE	Public Law 117-55, Secure Equipment Act of 2021, November 11, 2021.
Uyghur Forced Labor Prevention Act	2021	Ban imports of products produced in Xinjiang	Pub. L. 117-78, 2021.
CHIPS and Science Act and export restrictions on semiconductors	2022	Support the U.S. advanced semiconductor industry and prevent advanced semiconductor exports to China	Public Law 117-167, CHIPS and Science Act, August 9, 2022. Bureau of Industry and Security, 2022b.
Inflation Reduction Act	2022	Support U.S. clean energy technology and manufacturing	Pub. L. 117-169, 2022.

APPENDIX C

Timeline of Chinese Economic Policy Changes Following New U.S. Economic Policy Measures vis-à-vis China

Table C.1 summarizes Chinese economic policy changes that have followed the introduction of new U.S. economic policy measures.

TABLE C.1
Timeline of Chinese Economic Policy Changes Following New U.S. Economic Policy Measures vis-à-vis China

Measure	Date Implemented	Description
Investigation and countervailing duties	2018	China imposes 178.6 percent tariffs on sorghum imports from the United States.
First wave of retaliatory tariffs	2018	China imposes tariffs on aluminum scrap, autos, aircraft, and agricultural and other products worth $50 billion imported from the United States.
WTO filing	2018	China files WTO dispute, arguing U.S. tariffs have damaged China's trade interests.
Second wave of retaliatory tariffs	2018	China imposes tariffs on an additional $60 billion of goods imported from the United States.
Unreliable entity list	2019	China announces plans for its own "unreliable entity list."
Tariff rate increase	2019	China raises tariff rate on imports from the United States while lowering tariffs on imports from U.S. competitors.
National Integrated Circuit Industry Fund II	2019	China sets up $10 billion fund for semiconductor wafer fabrication projects.
Tariffs announced	2019	China threatens to impose tariffs on $75 billion of imports from the United States.
Trade agreement	2020	Under the Phase One trade agreement, China agrees to purchase $200 billion in imports from the United States.
14th Five-Year Plan	2021	China outlines various measures to improve its technology, food, and economic security.
Ban on U.S.-based chips	2023	China bans chips from U.S.-based Micron for use in information infrastructure.
National Integrated Circuit Industry Fund II	2023	Project is renewed with $29 billion in additional funding.

SOURCES: Features information from Chad P. Brown and Melina Kolb, "Trump's Trade War Timeline: An Up-to-Date Guide," Peterson Institute for International Economics, December 31, 2023; Etcetera Language Group, 2021; Ma Jingjing, "China's 'Big Fund II' Makes Intensive Investments, as Country Aims to Overcome US Chip Ban," *Global Times*, March 30, 2023; and Mullen, 2021.

Abbreviations

AI	artificial intelligence
ASML	Advanced Semiconductor Materials Lithography
BIL	Bipartisan Infrastructure Law
BIS	Bureau of Industry and Security
BRI	Belt and Road Initiative
BRICS	Brazil, Russia, India, China, and South Africa
CATL	Contemporary Amperex Technology Co., Limited
CFIUS	Committee on Foreign Investment in the United States
CHIPS	Creating Helpful Incentives to Produce Semiconductors
CIF	costs, insurance, and freight
COVID-19	coronavirus disease 2019
CPTPP	Comprehensive and Progressive Agreement for Trans-Pacific Partnership
DoD	Department of Defense
DOE	Department of Energy
DPA	Defense Production Act
EU	European Union
EV	electric vehicle
FDI	foreign direct investment
FIRRMA	Foreign Investment Risk Review Modernization Act
GDP	gross domestic product
IMEC	India–Middle East–Europe Corridor
IMF	International Monetary Fund
IPEF	Indo-Pacific Economic Framework for Prosperity
IRA	Inflation Reduction Act
NAFTA	North American Free Trade Agreement
PGII	Partnership for Global Infrastructure and Investment
R&D	research and development
SMIC	Semiconductor Manufacturing International Corporation
TPP	Trans-Pacific Partnership
WTO	World Trade Organization

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The authors identify two main goals of these recent policies: promoting fairer trade and defending U.S. economic interests. In their policy review, they find that U.S. economic policies achieved limited progress in promoting fairer trade but a higher degree of success in defending U.S. economic-related interests. Finally, the authors present several policy recommendations to better achieve these two goals related to trade, industry, controls on technology, economic diplomacy, foreign investment, and diversification of supply chains away from China.

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